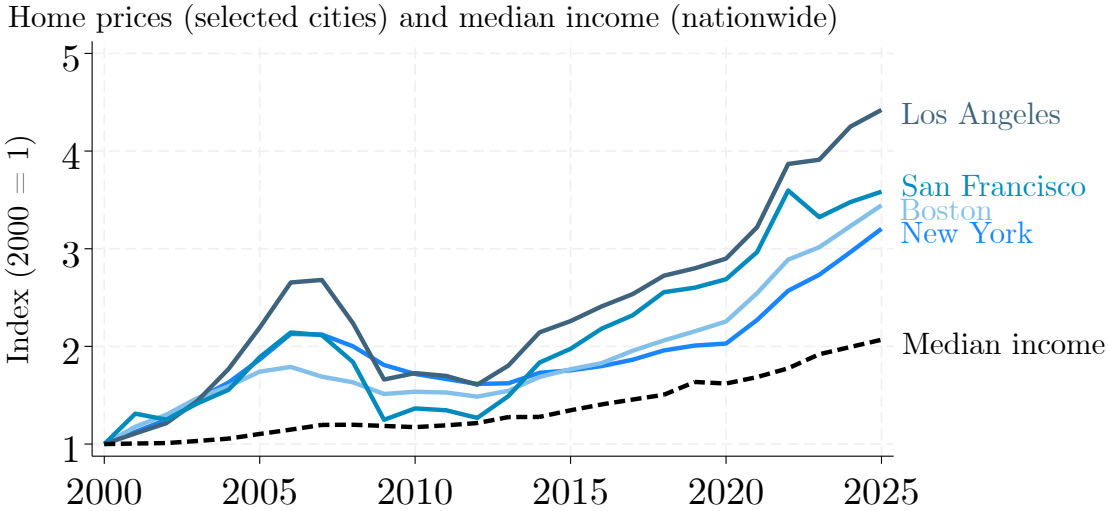
# Zoning and the Dynamics of Urban Redevelopment

**Vincent Rollet**

Stanford University

Boston College
September 10, 2026

# Home prices have skyrocketed in many large cities



# Strict zoning is often denounced as a culprit



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**Zoning**: Regulations restricting characteristics of new buildings.

- **Density limits** (e.g., height limits).
- **Use limits** (e.g., residential only).



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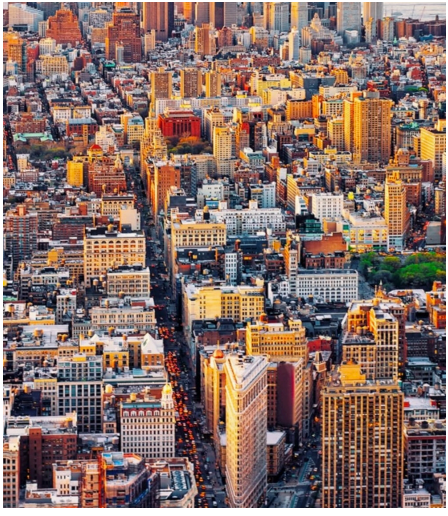
- **Density limits** (e.g., height limits).
- **Use limits** (e.g., residential only).

- Some deregulation (**upzoning**) happening in many cities (Chicago, Cambridge, Minneapolis, Berkeley, Denver, Seattle, Austin, NYC, Portland, Raleigh, San Diego...)



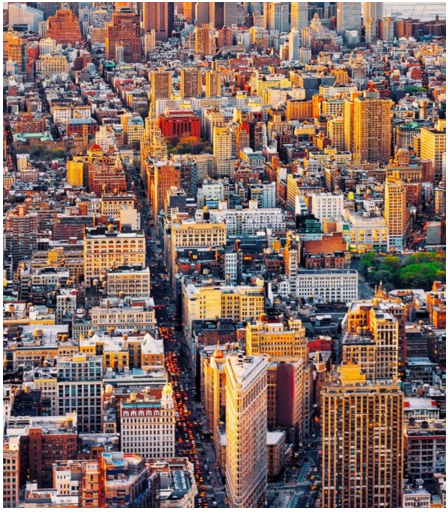
# Technological constraints matter too

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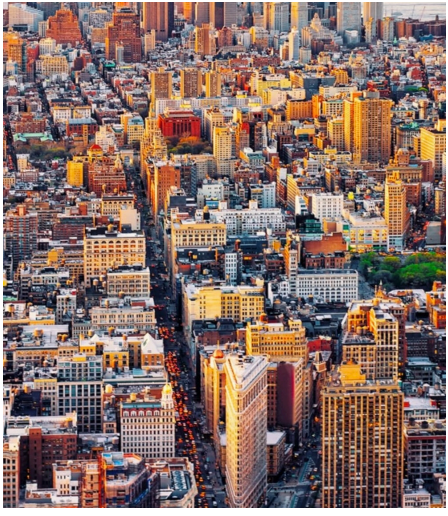
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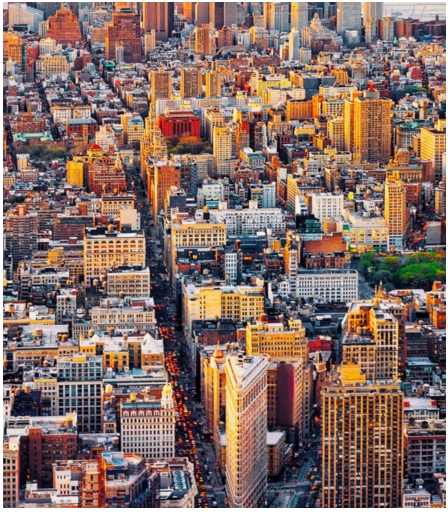
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- Change comes from **redevelopment**.
  → Demolishing and replacing old buildings.
- Increasingly how cities grow (Frolking et al., 2024).
- Requires paying large **fixed costs**.



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**Policy application: What would be the effects of relaxing zoning?**

- **Quasi-experimental evidence** from recent zoning changes.
- Model accurately predicts effects of past upzonings.
- Use model to predict the **effects of relaxing zoning** on **construction** and **prices**.

# Findings

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- **Consequences**: Relaxing zoning most effective where **prices are high**/**density is low**.
- Affordability **gains geographically spread out** due to migration.

# Contribution

**City structure and evolution** Burgess (1925), Hoyt (1939), Alonso (1964), Mills (1967), Muth (1969), Ogawa & Fujita (1980), Lucas & Rossi-Hansberg (2002), Ahlfeldt, Redding, Sturm & Wolf (2015), Allen, Arkolakis, & Li (2015), Heblich, Redding, & Sturm (2020), Gechter and Tsivanidis (2023), Hsiao (2024)

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→ Evaluation of the **effects of relaxing zoning** (reduced-form + credible extrapolation).

# Roadmap

1. Data and context
2. Empirical evidence on redevelopment and zoning
3. Dynamic model of urban change
4. Model estimation and validation
5. Implications for zoning policy

# Data and context

# NYC as a case study

- Largest US city (8.5 million inhabitants).



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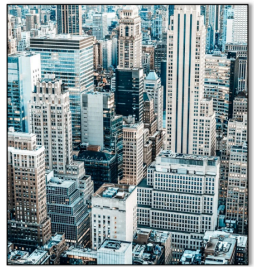
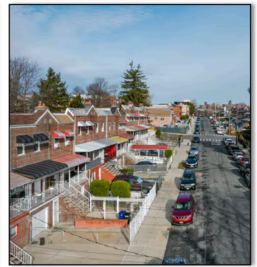
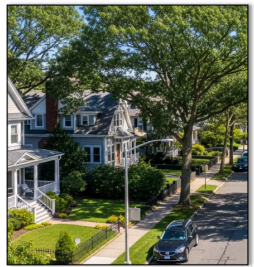
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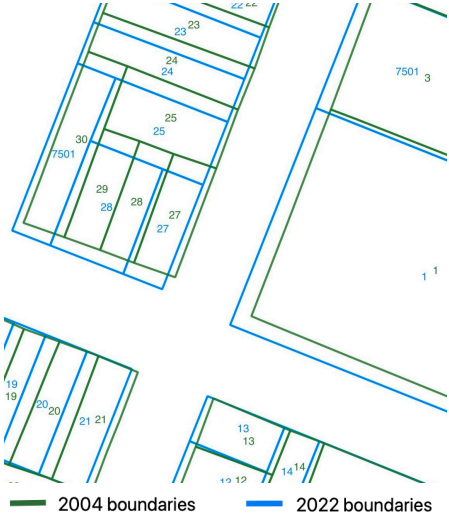
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- Need to gather data on:
  1. Floorspace quantities.
  2. Floorspace prices.
  3. Zoning.

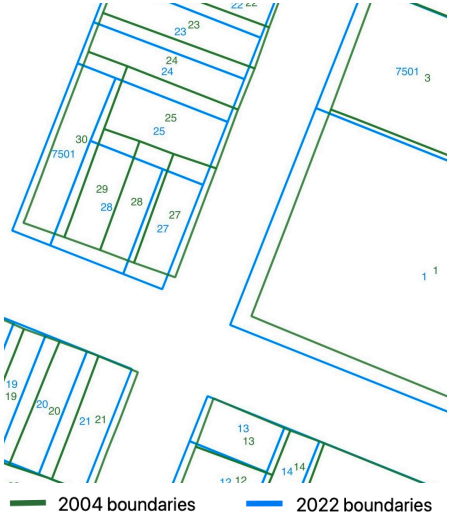

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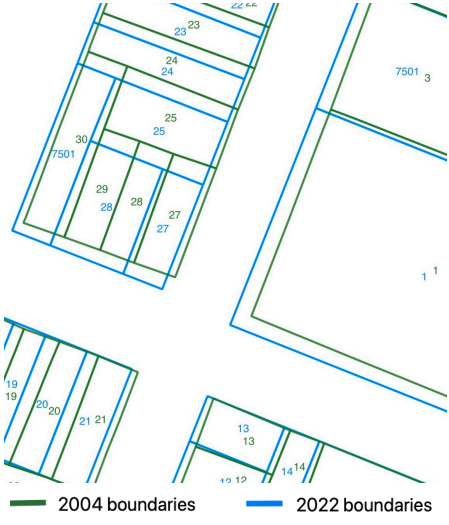
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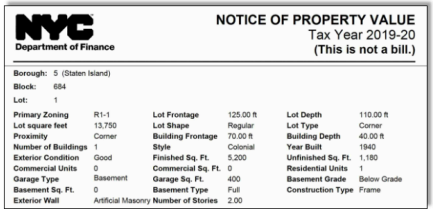
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- **Building characteristics**:
  - → Property tax records (scraping, FOIA requests).
  - → StreetEasy (scraped data on 1.6 million units).

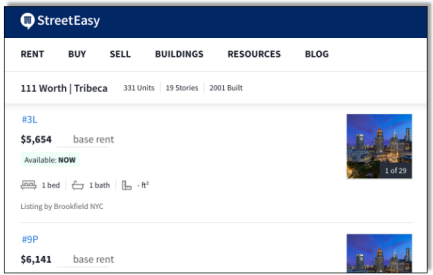


NYC Department of Finance

NOTICE OF PROPERTY VALUE
Tax Year 2019-20
(This is not a bill.)

Borough: 5 (Staten Island)
Block: 684
Lot: 1

| | | | | | |
|---|---|---|---|---|---|
| Primary Zoning | R1-1 | Lot Frontage | 125.00 ft | Lot Depth | 110.00 ft |
| Lot square feet | 13,750 | Lot Shape | Regular | Lot Type | Corner |
| Proximity | Corner | Building Frontage | 70.00 ft | Building Depth | 40.00 ft |
| Number of Buildings | 1 | Style | Colonial | Year Built | 1940 |
| Exterior Condition | Good | Finished Sq. Ft. | 5,200 | Unfinished Sq. Ft. | 1,180 |
| Commercial Units | 0 | Commercial Sq. Ft. | 0 | Residential Units | 1 |
| Garage Type | Basement | Garage Sq. Ft. | 400 | Basement Grade | Below Grade |
| Basement Sq. Ft. | 0 | Basement Type | Full | Construction Type | Frame |
| Exterior Wall | Artificial Masonry | Number of Stories | 2.00 | | |



StreetEasy

RENT BUY SELL BUILDINGS RESOURCES BLOG

111 Worth | Tribeca 331 Units | 19 Stories | 2001 Built

#3L
$5,654 base rent
Available: NOW
1 bed | 1 bath | · ft²
Listing by Brookfield NYC
1 of 29

#9P
$6,141 base rent

Boundary changes

# Floorspace quantities: Redevelopment events

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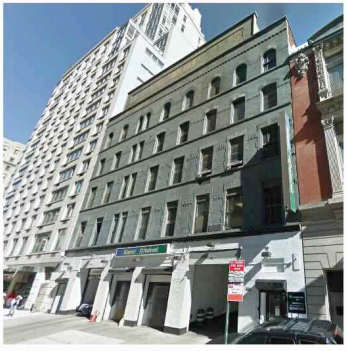
- **Building permits**

# Floorspace quantities: Redevelopment events

- Isolate 22,000 **redevelopment events**.
  → Buildings constructed since 2004.
- **Building permits**
- **Certificates of Occupancy**
  → Construction ends.
  → Scraped and digitized 250,000 documents.



# Floorspace quantities: Redevelopment timelines







2014 — 2015 — 2017 →

Building permit issued

Certificate of Occupancy issued

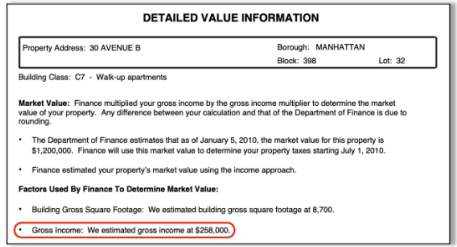
# Floorspace prices

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  - → From NYC Department of Finance.

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- Yearly **rent** measures for 73,000 buildings.
  → Scraped 600,000 tax documents.



**DETAILED VALUE INFORMATION**

Property Address: 30 AVENUE B
Borough: MANHATTAN
Block: 398 Lot: 32

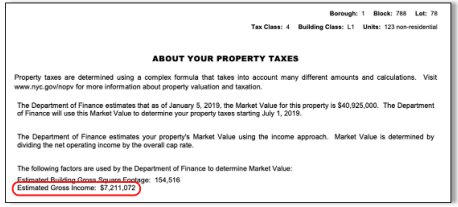
Building Class: C7 - Walk-up apartments

**Market Value:** Finance multiplied your gross income by the gross income multiplier to determine the market value of your property. Any difference between your calculation and that of the Department of Finance is due to rounding.

- The Department of Finance estimates that as of January 5, 2010, the market value for this property is $1,200,000. Finance will use this market value to determine your property taxes starting July 1, 2010.
- Finance estimated your property's market value using the income approach.

**Factors Used By Finance To Determine Market Value:**

- Building Gross Square Footage: We estimated building gross square footage at 8,700.
- Gross Income: We estimated gross income at $258,000.



**Borough:** 1 **Block:** 788 **Lot:** 78
**Tax Class:** 4 **Building Class:** L1 **Units:** 123 non-residential

**ABOUT YOUR PROPERTY TAXES**

Property taxes are determined using a complex formula that takes into account many different amounts and calculations. Visit www.nyc.gov/nopv for more information about property valuation and taxation.

The Department of Finance estimates that as of January 5, 2019, the Market Value for this property is $40,925,000. The Department of Finance will use this Market Value to determine your property taxes starting July 1, 2019.

The Department of Finance estimates your property's Market Value using the income approach. Market Value is determined by dividing the net operating income by the overall cap rate.

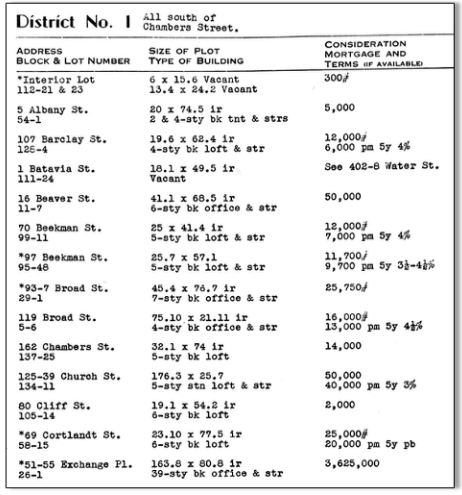
The following factors are used by the Department of Finance to determine Market Value:
Estimated Building Gross Square Footage: 154,516
Estimated Gross Income: $7,211,072

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- Digitized historical sales data from the Real Estate Board of New York.
  → 63,000 sales over 1950–1985.



**District No. 1** All south of Chambers Street.

| ADDRESS BLOCK & LOT NUMBER | SIZE OF PLOT TYPE OF BUILDING | CONSIDERATION MORTGAGE AND TERMS (IF AVAILABLE) |
|---|---|---|
| *Interior Lot 112-21 & 23 | 6 x 15.6 Vacant 13.4 x 24.2 Vacant | 300# |
| 5 Albany St. 54-1 | 20 x 74.5 ir 2 & 4-sty bk tnt & strs | 5,000 |
| 107 Barclay St. 128-4 | 19.6 x 62.4 ir 4-sty bk loft & str | 12,000# 6,000 pm 5y 4% |
| 1 Batavia St. 111-24 | 18.1 x 49.5 ir Vacant | See 402-8 Water St. |
| 16 Beaver St. 11-7 | 41.1 x 68.5 ir 6-sty bk office & str | 50,000 |
| 70 Beekman St. 99-11 | 25 x 41.4 ir 5-sty bk loft & str | 12,000# 7,000 pm 5y 4% |
| *97 Beekman St. 95-48 | 25.7 x 57.1 5-sty bk loft & str | 11,700# 9,700 pm 5y 3½-4½% |
| *93-7 Broad St. 29-1 | 45.4 x 76.7 ir 7-sty bk office & str | 25,750# |
| 119 Broad St. 5-6 | 75.10 x 21.11 ir 4-sty bk office & str | 16,000# 13,000 pm 5y 4½% |
| 162 Chambers St. 137-25 | 32.1 x 74 ir 5-sty bk loft | 14,000 |
| 125-39 Church St. 134-11 | 176.3 x 25.7 5-sty stn loft & str | 50,000 40,000 pm 5y 3% |
| 80 Cliff St. 105-14 | 19.1 x 54.2 ir 6-sty bk loft | 2,000 |
| *69 Cortlandt St. 58-15 | 23.10 x 77.5 ir 6-sty bk loft | 25,000# 20,000 pm 5y pb |
| *51-55 Exchange Pl. 26-1 | 163.8 x 80.8 ir 39-sty bk office & str | 3,625,000 |

# Zoning

- I match my land use panel with zoning regulations for each year.

# Zoning

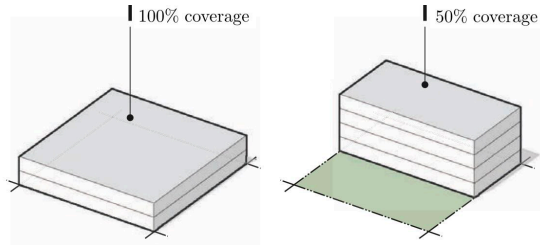
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# Zoning

- I match my land use panel with zoning regulations for each year.
- Zoning limits the **size** and **use** (residential vs. commercial) of new buildings.
- Main zoning instrument in NYC: limits on the **Floor Area Ratio** (FAR).

$$\textbf{FAR} = \frac{\text{sq. ft of floorspace}}{\text{sq. ft of land}}$$

**Buildings with a FAR of 2**



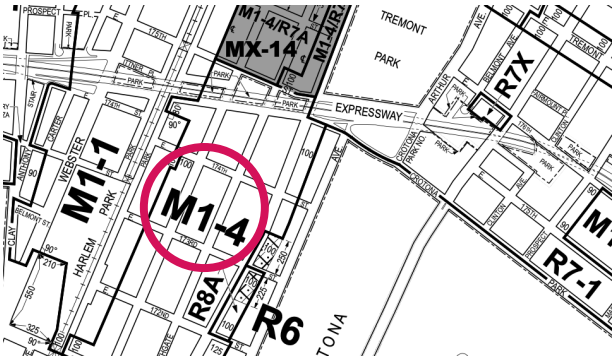

R3X
R1-2
R3A
R3-1
R3-1
R3-2
PARK
GREAT KILLS
KILLS
RD.
GREEN
HOLTON
ADRIENNE
CT.
PL.
DENT
RUSTIC
RUSTIC PL.
FELDWAY
TRENT
DURANT
MAYBURY
GREENCROFT
AVE.
WATERSIDE PKWY.
FAIRLAWN
BLVD.
ST.
KEATS
GORMAN
VOLGA
VOLGA AVE.
250
180
170
150
S.S.L. OF R.R. ROW




## Zone R1-2

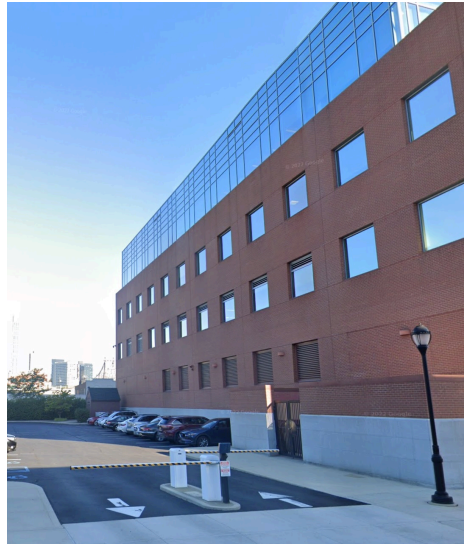
- **Residential** allowed up to 0.5 FAR points.
- **Commercial** not allowed.

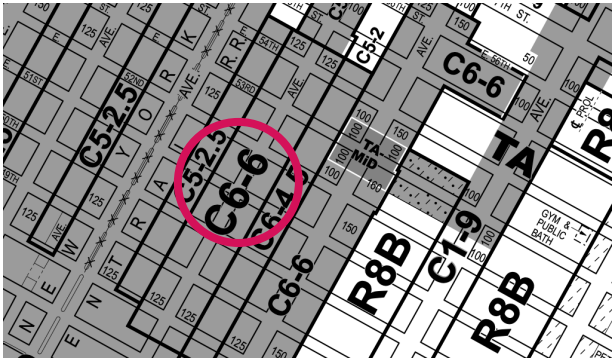





## Zone M1-4

- **Residential** not allowed.
- **Commercial** allowed up to 2 FAR points.









## Zone C6-6

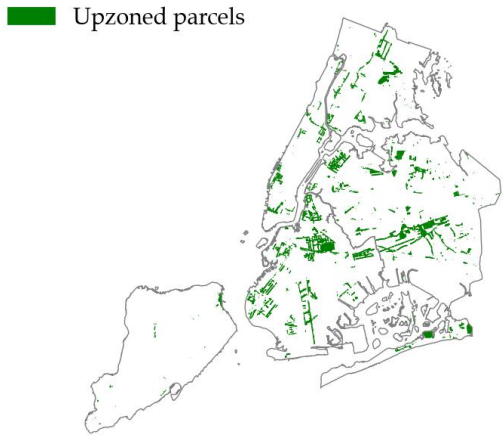
- **Residential** allowed up to 10 FAR points.
- **Commercial** allowed up to 15 FAR points.

- Zoning ordinance adopted in 1961.
  - → Promoted “towers in the park.”
  - → Separated residential/commercial uses.
  - → Goal was to limit negative externalities.

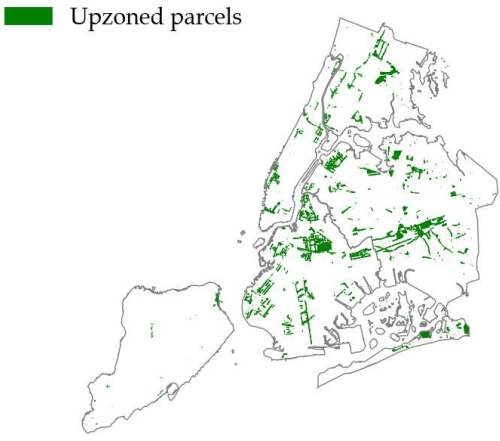
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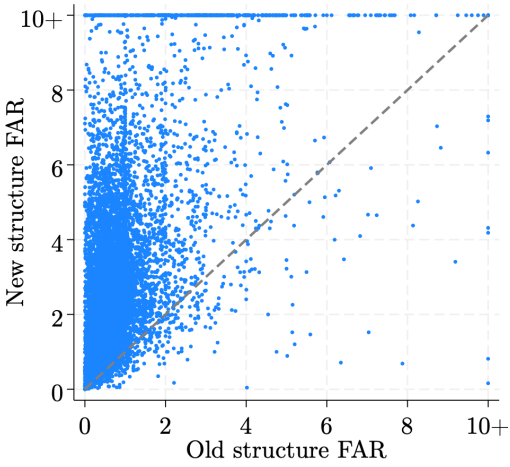
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- Strong persistence over time.
- Several areas upzoned in past decades.
  - → Allows to evaluate the effects of upzoning.
- Focus on large planner-initiated changes.
  - → Many veto players.
  - → Exact changes and timing unclear when upzoning discussions begin.
  - → Leveraged in reduced-form.



# Empirical evidence

# Redevelopment means densification

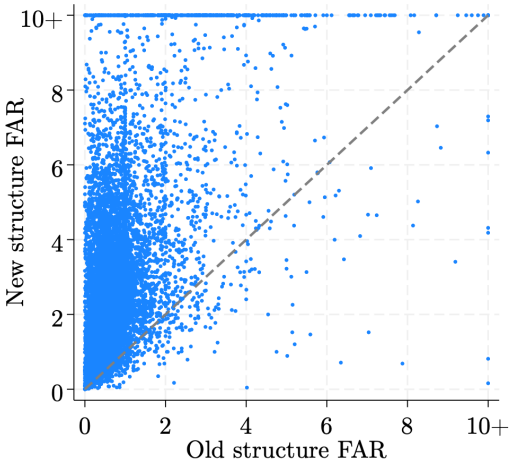
- New buildings 3.4 times larger than the ones they replace (on average).




Change in number of units

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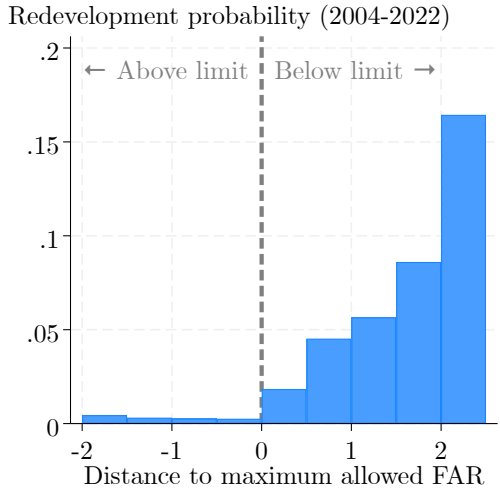
- New buildings 3.4 times larger than the ones they replace (on average).
- Suggests large redevelopment **fixed costs**.




Change in number of units

# Redevelopment mostly happens when upward growth is allowed

- Buildings at/over the zoning limit are seldom redeveloped.



# The effects of upzoning on built FAR materialize slowly

- Compare parcels upzoned earlier vs. later.
  → Identifying assumption: Exogenous timing.
  (de Chaisemartin & d'Haultfœuille, 2020)

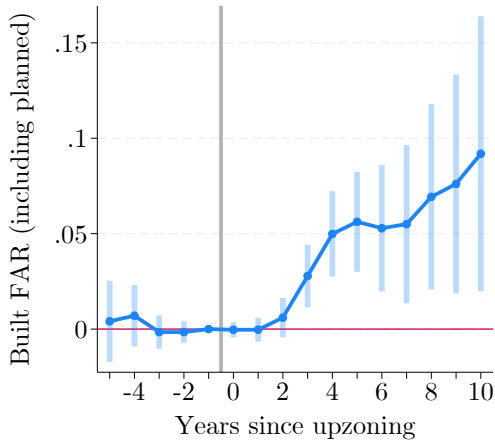
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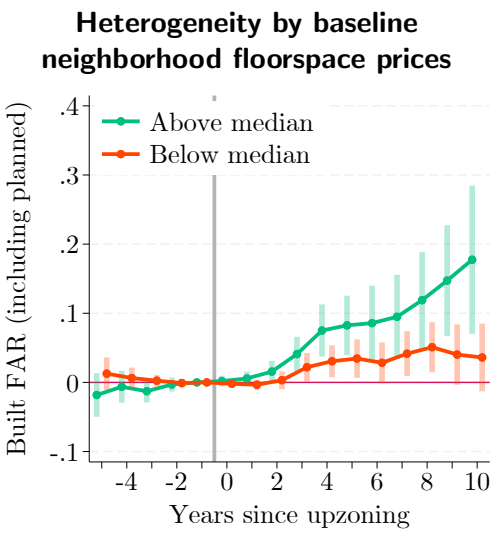
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- $\approx$ 10% of increase used after a decade.




Relaxing use constraints

# The effects of upzoning vary widely

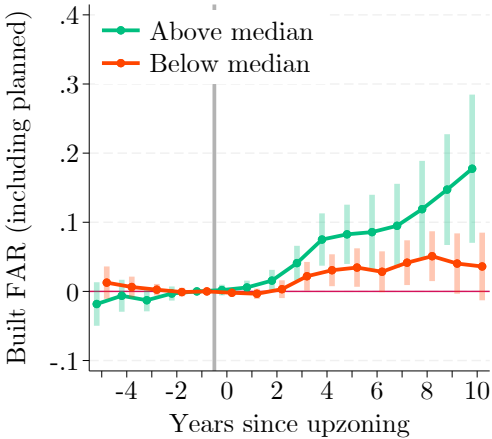
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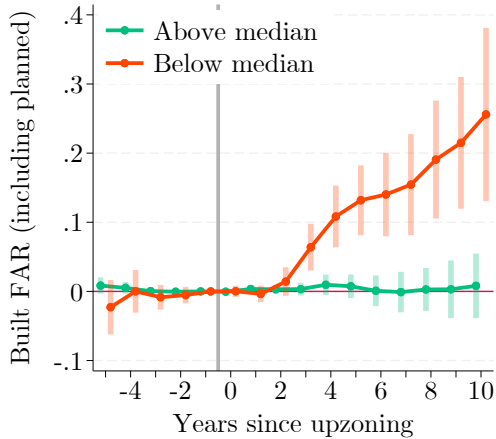
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**Heterogeneity by baseline neighborhood floorspace prices**

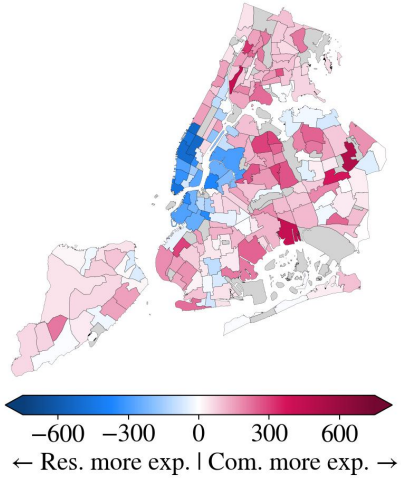



**Heterogeneity by baseline built FAR (within neighborhood)**



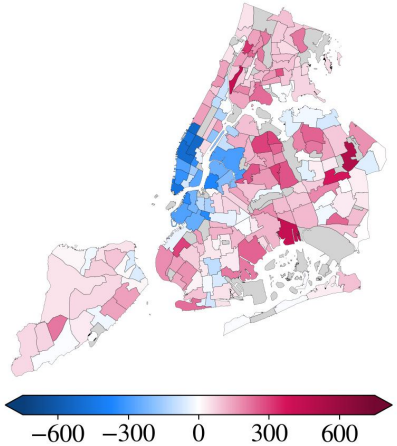
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Commercial-residential price diff. ($/sq. ft)

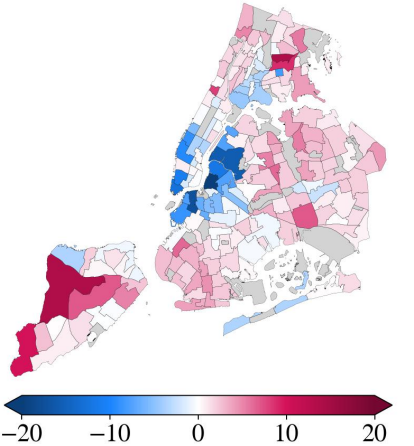


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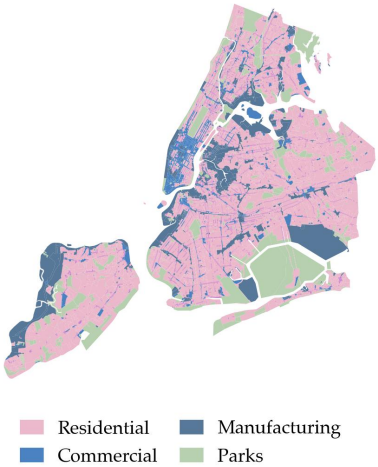



2004-2022 change in commercial share (pp)

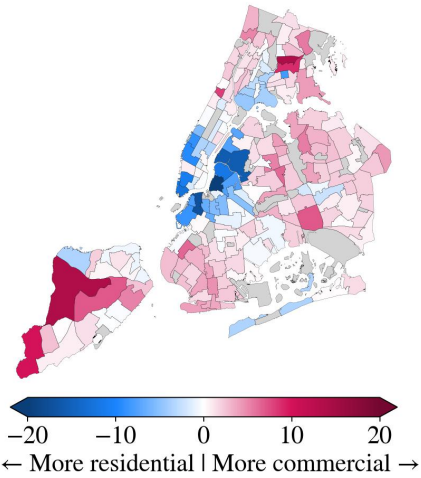


# Zoning constrains the reallocation of land uses

Zoning map




2004-2022 change in commercial share (pp)



# Remaining questions

1. **Extrapolation**:
   - → Upzoned neighborhoods were selected: what would happen if we upzoned **elsewhere**?
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→ Use reduced-form analysis of upzoning for **model validation**.

# A dynamic model of urban change

# Supply of floorspace

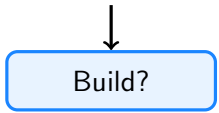
Dynamic behavior of developers given prices and regulation

# Developer choices

Developers are atomistic and perfectly foresighted. They make decisions each year:

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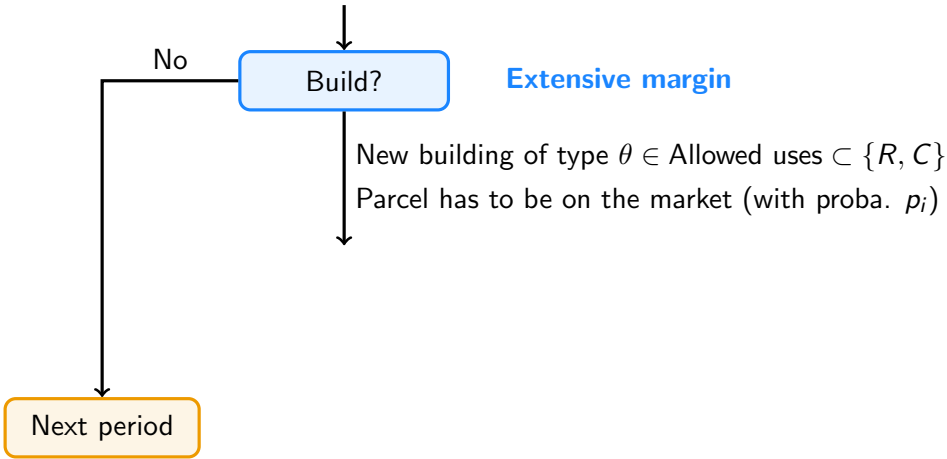
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**Extensive margin**

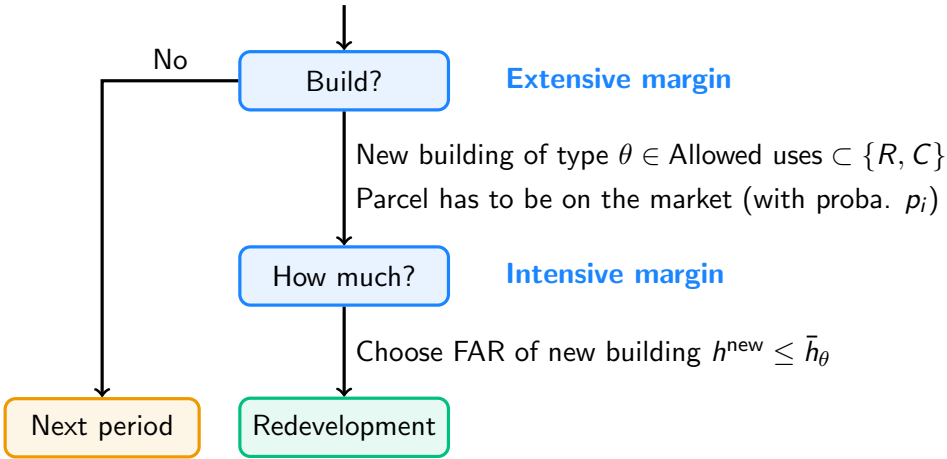
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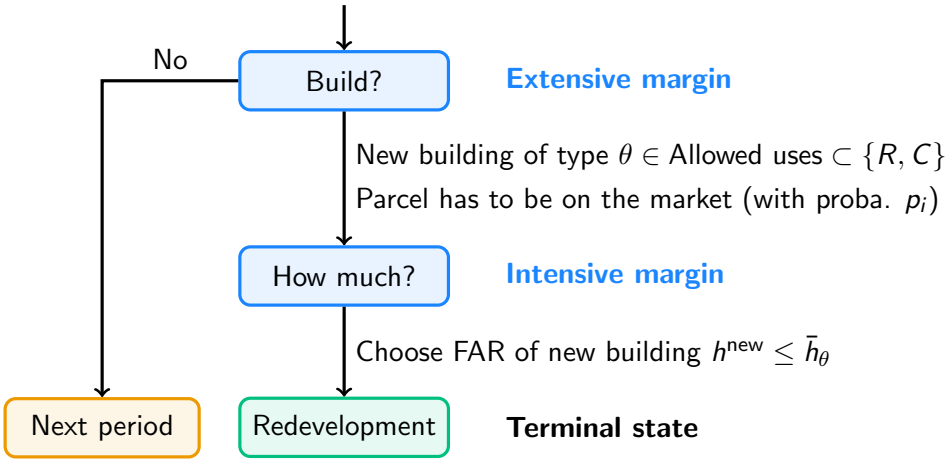
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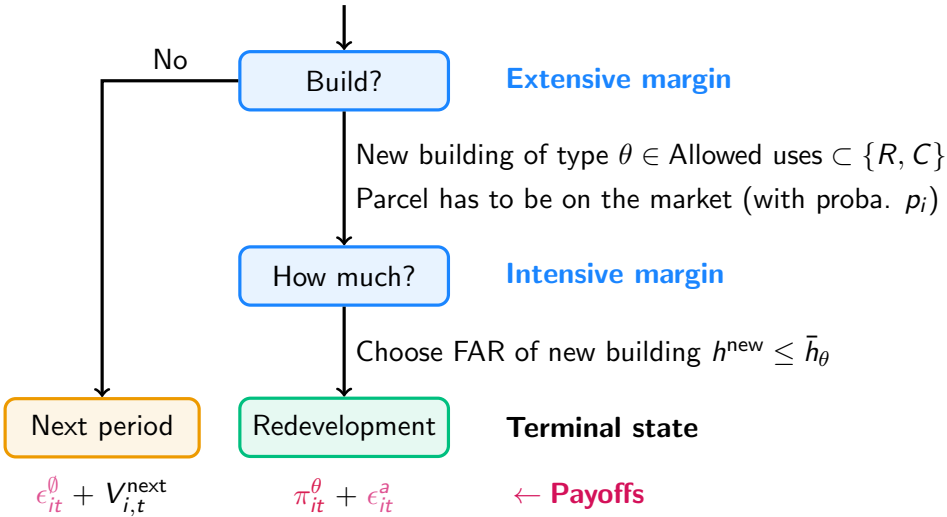
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- Fixed costs ($FC$): eviction, demolition, permitting, etc.
  $\rightarrow$ Vary by building (e.g., larger for bigger buildings in dense neighborhoods).

# Demand for floorspace

Evolution of prices given developers' choices

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$$\max_{n,m,H,C} \underbrace{B_{nt}}_{\text{amenities}} \cdot \underbrace{(H - \underline{H}_{nt})^{\beta}}_{\text{housing}} \cdot \underbrace{C^{1-\beta}}_{\text{other cons.}} \cdot \underbrace{d_{nmt}^{-1}}_{\text{commuting}} \cdot \underbrace{z_{nt}^{H} z_{mt}^{W}}_{\substack{\text{idiosyncratic} \\ \text{preferences for} \\ \text{home/work locations}}}$$

$$\text{s.t. } C + R_{nt} H \leq \text{Income}_t$$

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- **Key extension**: Subsistence level of housing and realistic distribution of worker skills.
  → Essential to rationalize consumed quantities of housing.

# Amenities and productivities

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- City population and congestion increase with expected utility.

# Dynamic spatial equilibrium

An equilibrium is a path of

- → Local floorspace supplies, rents, and prices;
- → Population and job distributions;
- → Local productivities, amenities, wages;
- → Commuting costs;

Such that:

1. Developers maximize profits.
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# Model estimation and validation

# Supply-side parameters

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# Prices

- Value of a building:

$$P_{\theta it} = \underbrace{\rho_{nt}^{\theta}}_{\text{Location premium}} \times \underbrace{Q(\mathbf{x}_{it})}_{\text{Building quality}} \times \underbrace{h_{it}}_{\text{Building size}}$$

→ Estimated through a hedonic regression on the sales data.
→ Focus on buildings at/above FAR limit (option value of redevelopment $\approx 0$).
→ Quality proxied using rich building characteristics in $\mathbf{x}_{it}$ (e.g., age, grade, materials, FAR).

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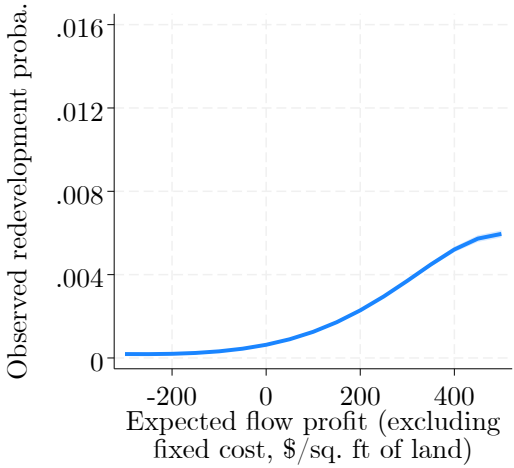
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- I find costs rise steeply with FAR and align with engineering estimates/reported costs.

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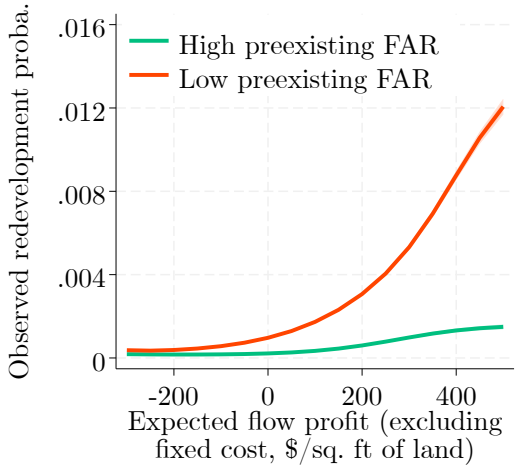
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- **Parameterization**: fixed costs
  - → Increase with size of old building.
  - → Increase with neighborhood density.
  - → Larger in historic districts.



# Fixed costs estimation: Algorithm

- Requires solving the full model.
  - → Option value of redevelopment depends on future path of prices.
  - → This path depends on the collective behavior of developers, which depends on fixed costs.
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- I find fixed costs increase sharply with existing buildings' FAR.

# Demand-side parameters

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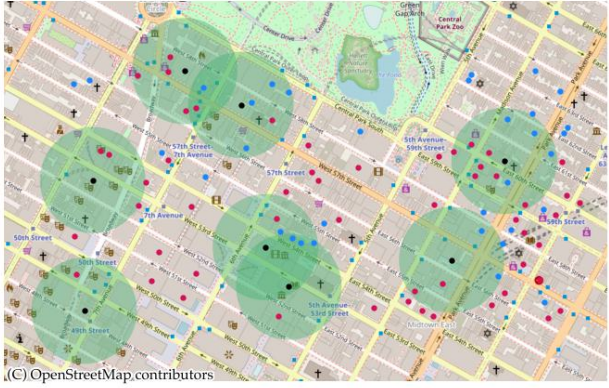
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# Price effects of new construction

- Draw 500-ft disks around large construction events (290 residential, 112 commercial).
- Compare the evolution of rents in disks treated earlier vs. later.




**Events (new residential buildings)**

New buildings

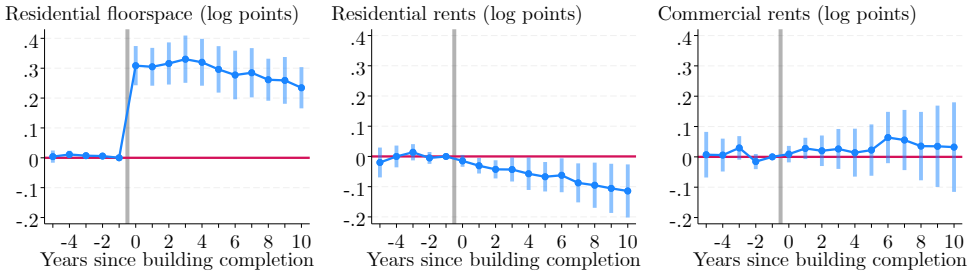
500 ft buffers

**Buildings with available rent data**

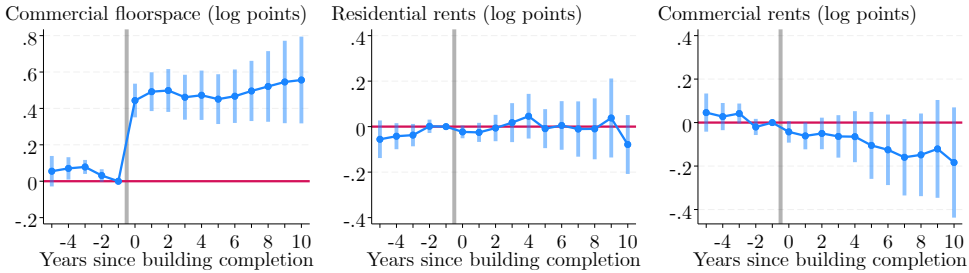
Residential

Commercial

## Effects of residential construction




## Effects of commercial construction



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- I find positive agglomeration externalities within uses, limited externalities across uses.
  - → Effect of residents on other residents: $\gamma_{RR} = 0.11$.
  - → Effect of firms on other firms: $\gamma_{CC} = 0.07$.
  - → Corresponding estimates in Ahlfeldt et al. (2015): 0.16 and 0.07.

# Model validation

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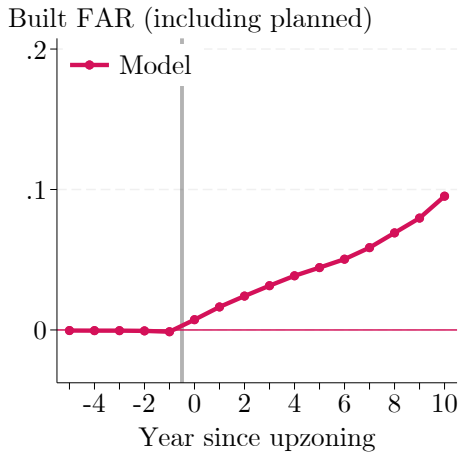
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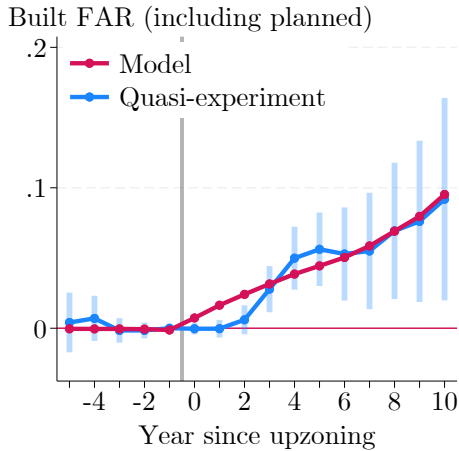
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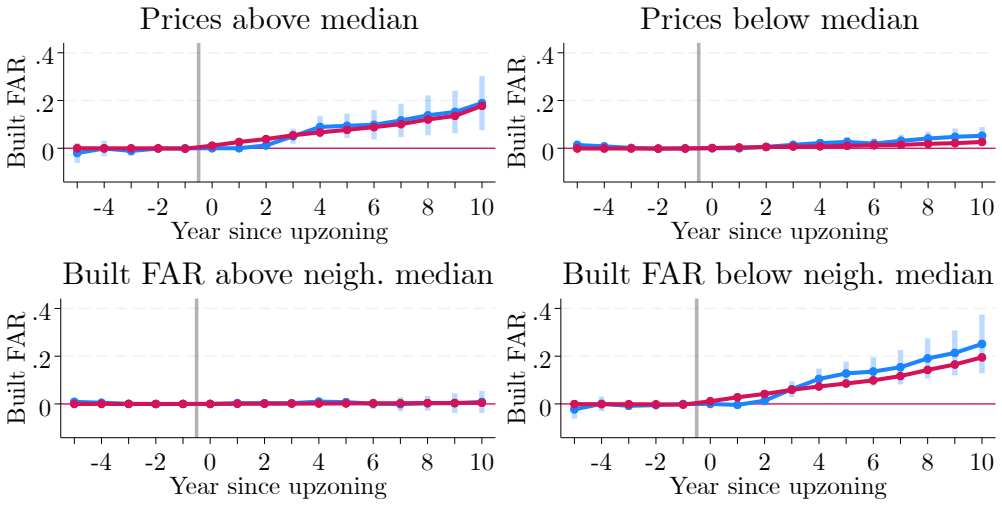
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# Model validation: Predicted effect of upzonings



# Implications for zoning policy

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Upzoning map | Protected parcels | Flood zones

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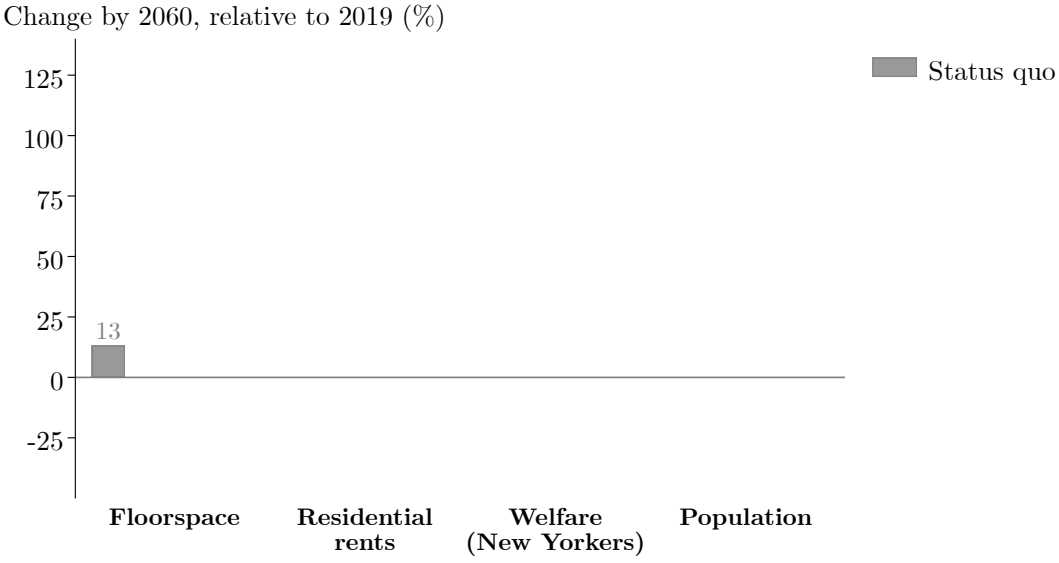
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4. **Frictionless benchmark** (no zoning, no adjustment costs, price = marginal cost).

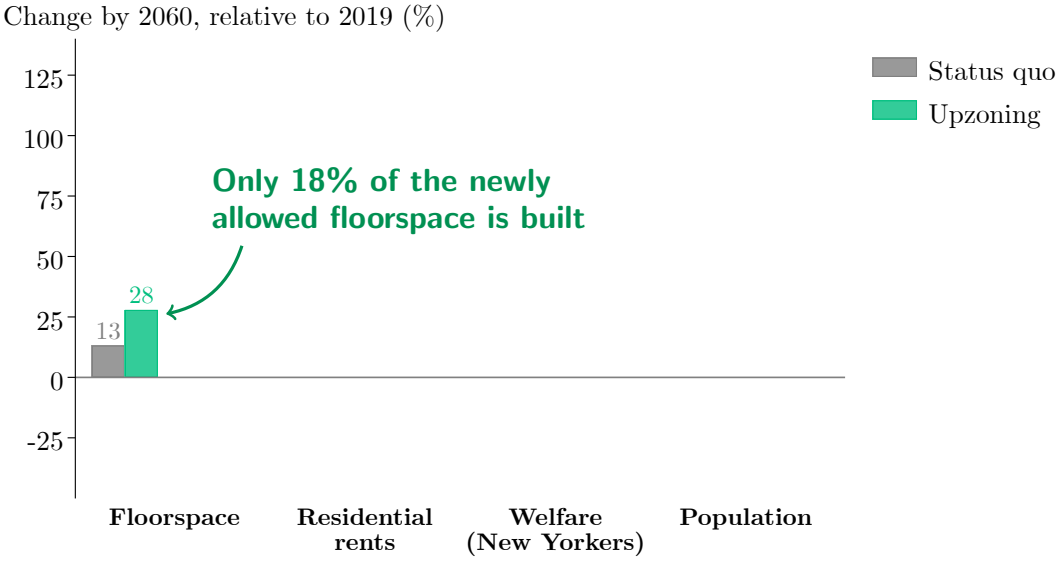
Upzoning map | Protected parcels | Flood zones

# To what extent does zoning constrain NYC's growth?

# Under current zoning, NYC continues to grow slowly



# Upzoning increases floorspace by 15pp over 40 years



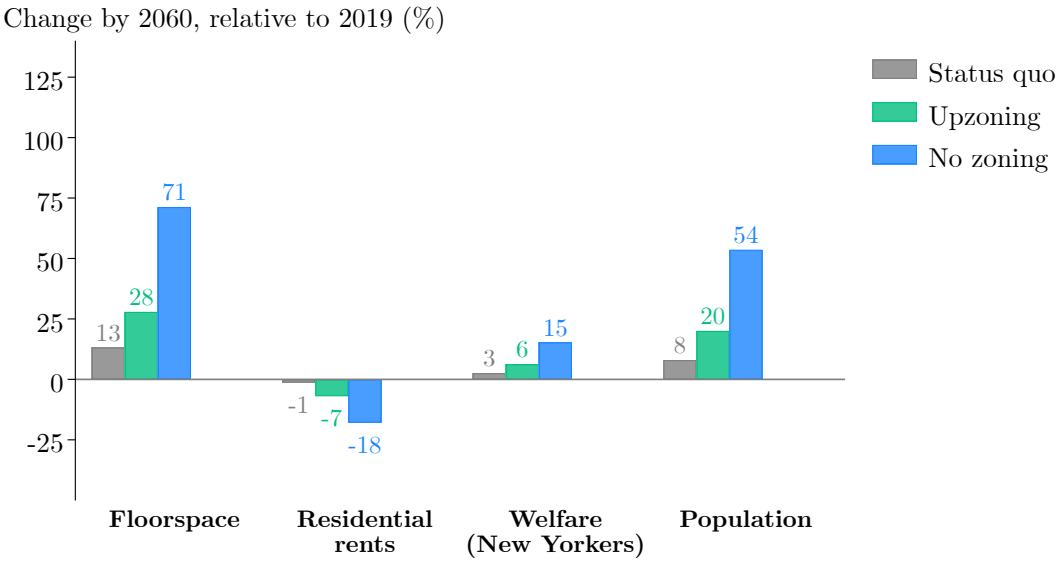

Effects materialize slowly over time

# Completely removing zoning quintuples NYC's growth rate

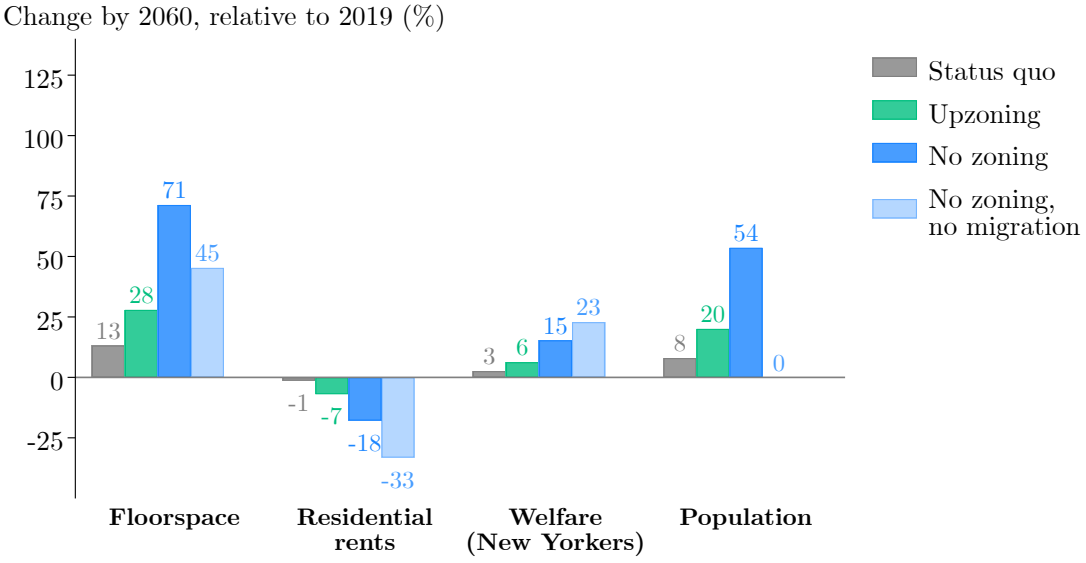



Separately relaxing FAR and use limits

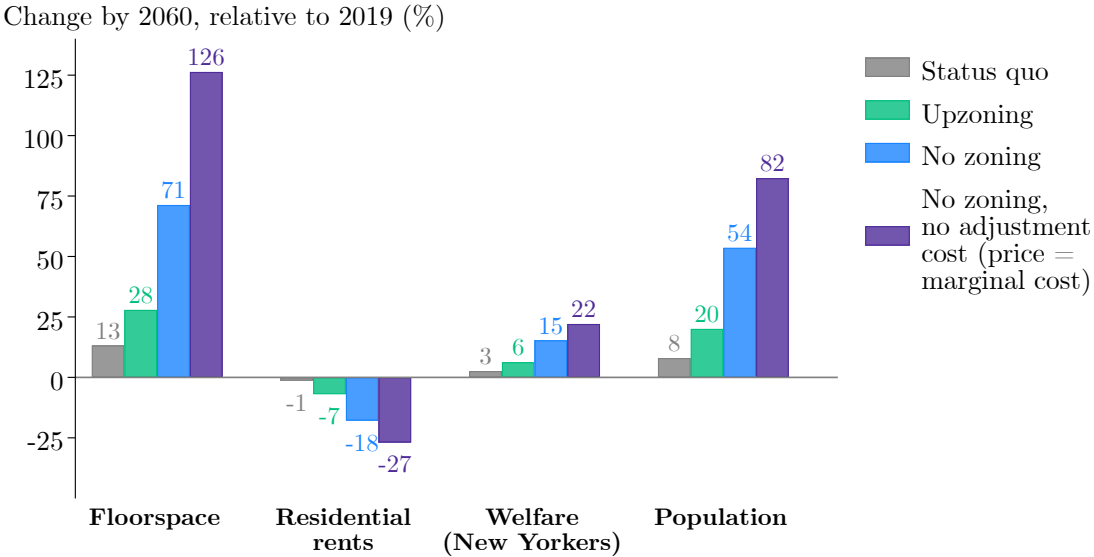
# Rent decreases are moderate



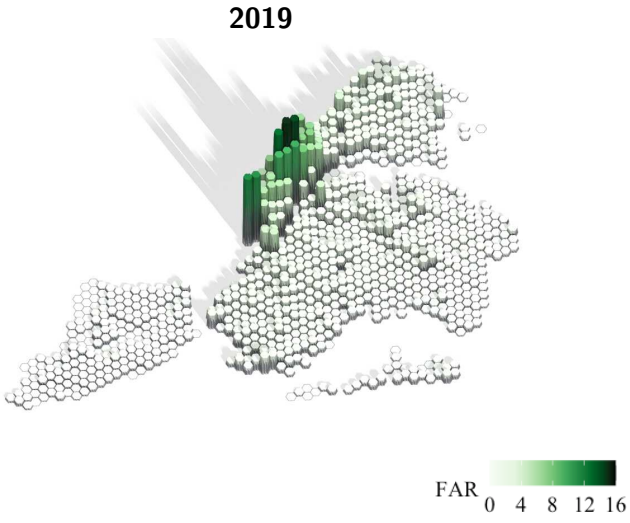
# Rent decreases are moderated by migration



# A model ignoring redevelopment greatly overstates effects



# Technological constraints matter for credible counterfactuals

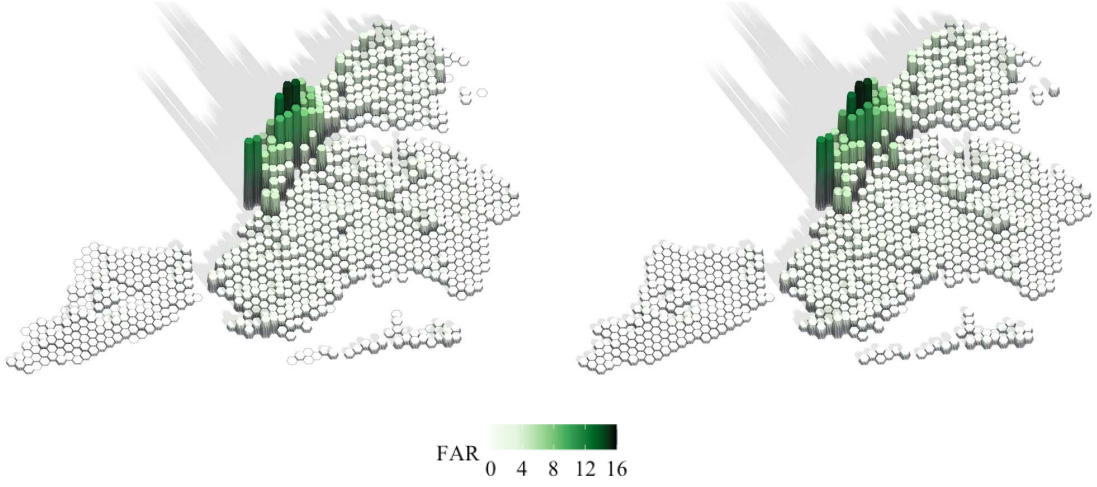



More

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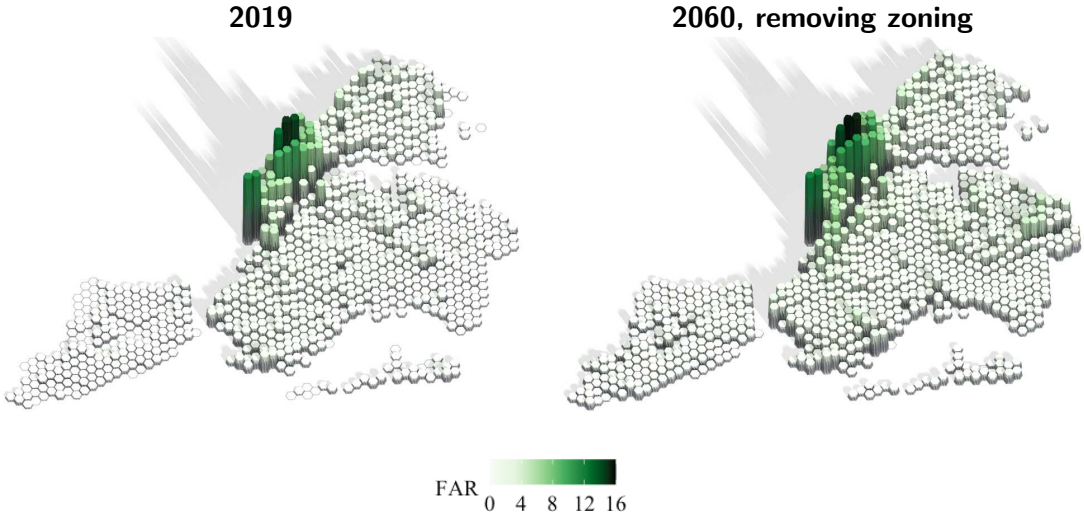
**2019**

**2060, status quo zoning**




More

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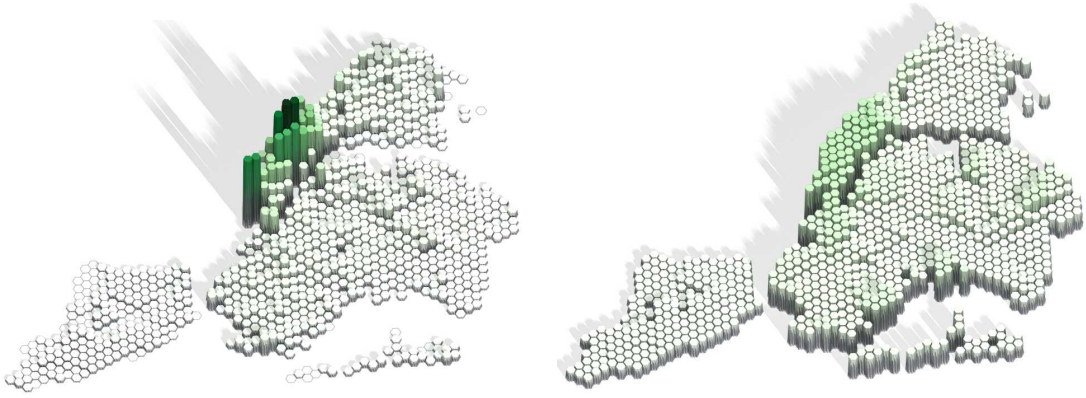



More

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**2019**

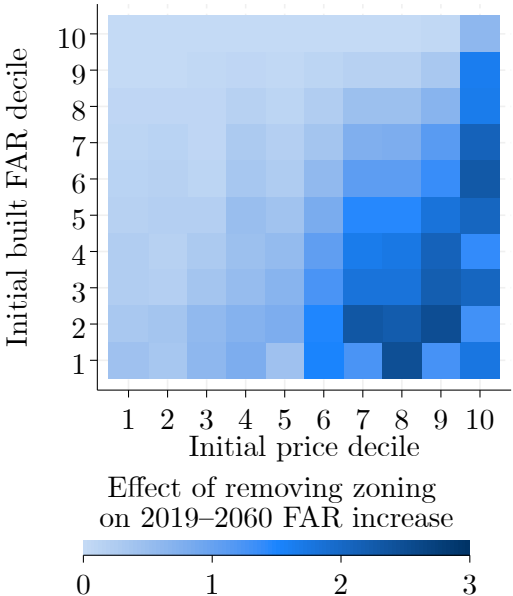
**No zoning, no adjustment costs**



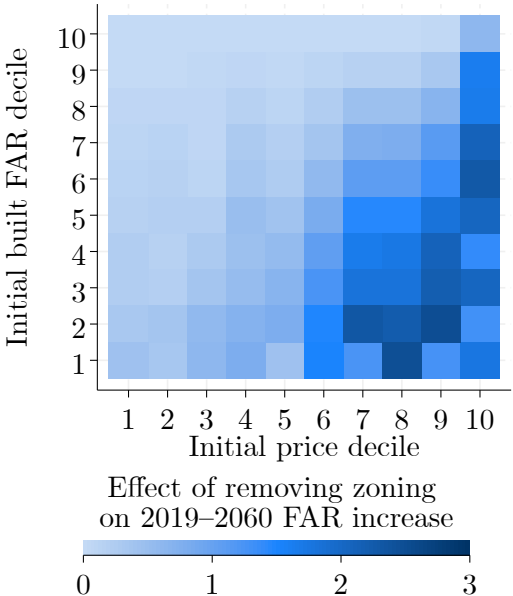

More

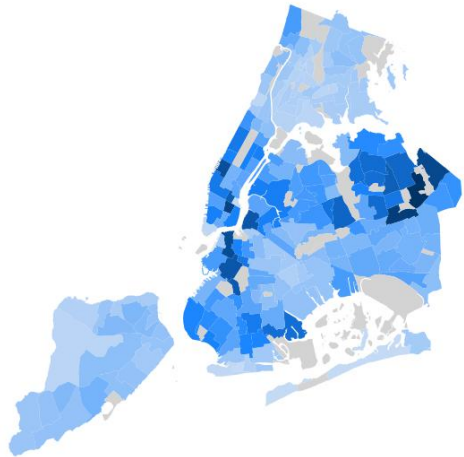
# Where is zoning a constraint?

# Where does relaxing zoning lead to increased supply?

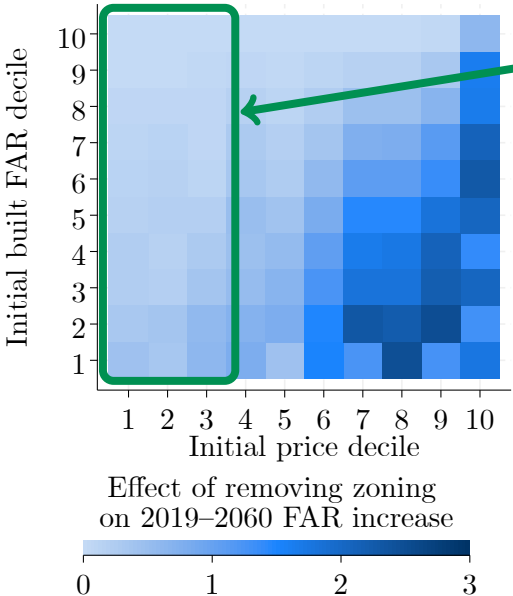


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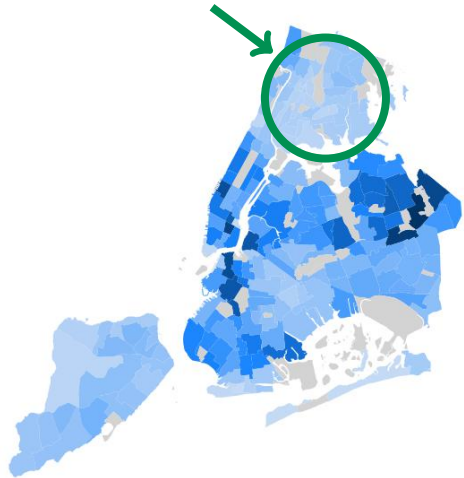


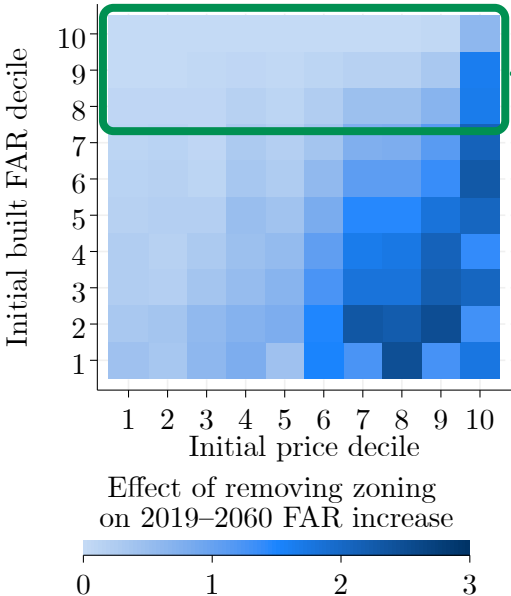
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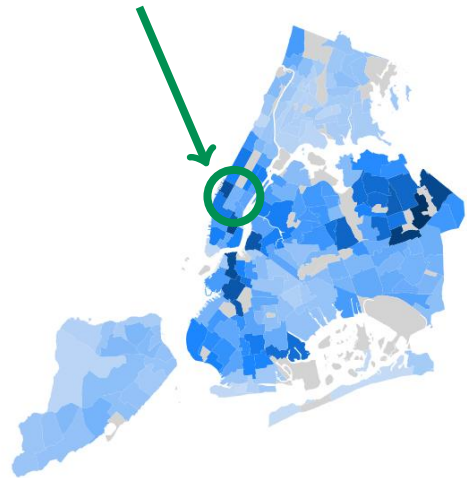

Zoning tends not to be binding when floorspace prices are low.



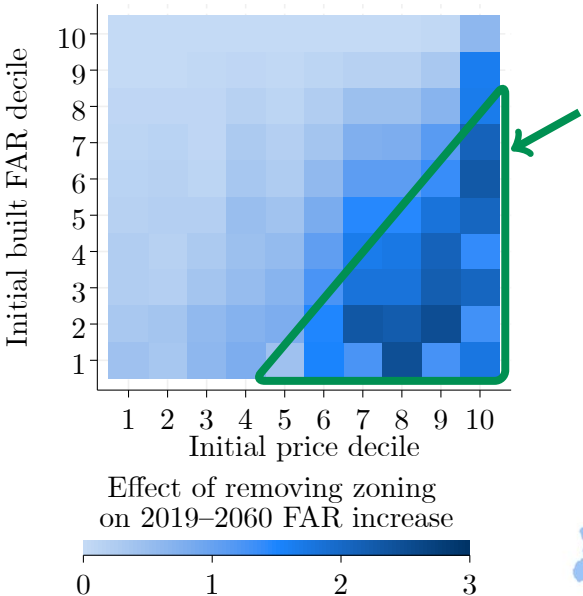
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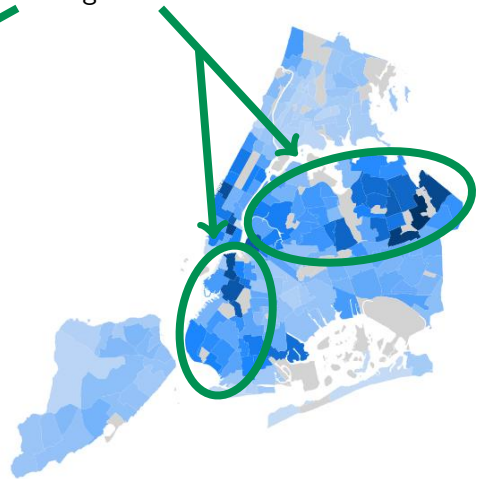

Removing zoning has little effect on parcels with tall buildings.



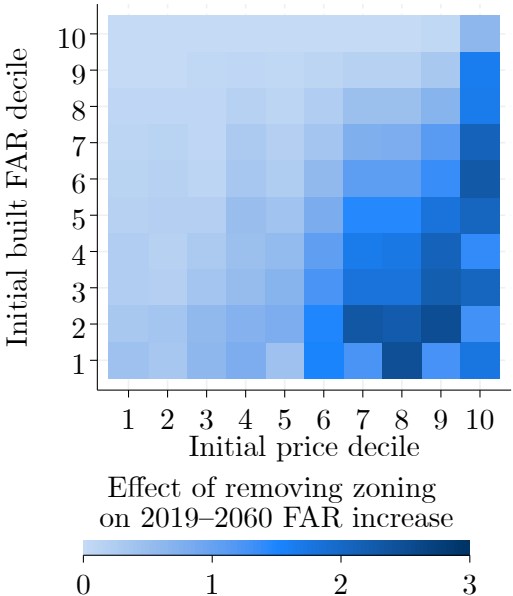
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Upzoning is effective when prices are high enough and density is low enough.

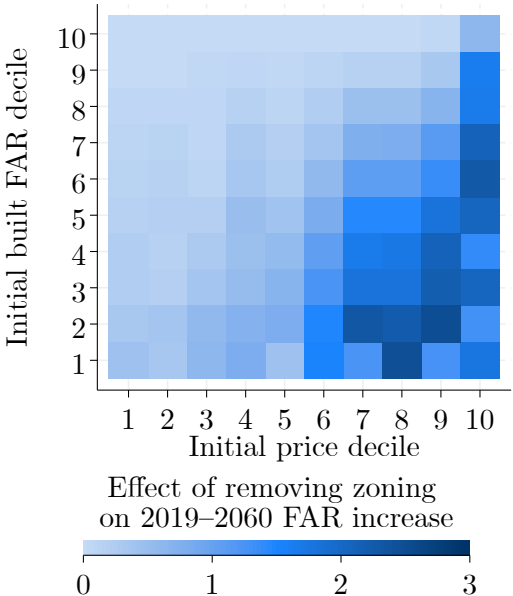


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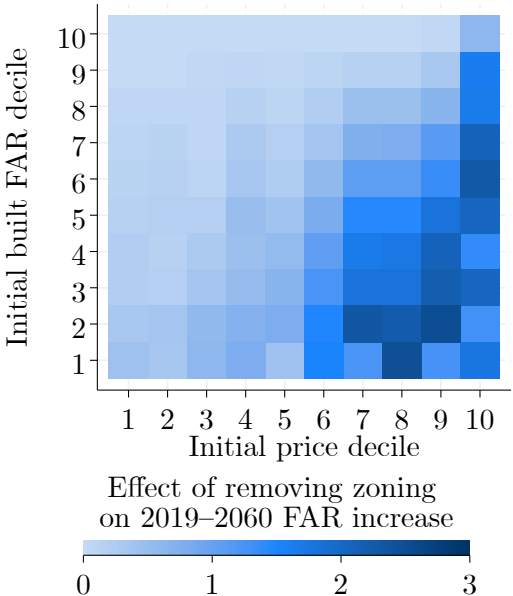

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- NYC's planners already do targeting.
- Upzoning would likely have limited effects in inexpensive cities.

# Additional results

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- **Distributional effects**. More
  - → Lower-income workers benefit more.

- **Alternative policies** to promote construction and affordability. More
  - → Upzoning outperforms reducing construction costs and tax incentives.
  - → Contribution to the literature on the topic: Glaeser and Gyourko (2008), Soltas (2024)...

- **Supply elasticities**. More
  - → Compute supply elasticities at a granular level and different time horizons.
  - → Supply elasticities vary widely across neighborhoods. Zoning is their primary determinant.
  - → Contribution to the literature on the topic: Saiz (2010), Baum-Snow and Han (2024)...

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- Much has changed since 1961:
  - → Floorspace prices have skyrocketed, manufacturing activity has plummeted.
- But zoning has been much more persistent than planners anticipated and intended.
  - → Rezoning is vulnerable to obstruction by those with a stake in the status quo.

Historical population and prics | Manufacturing externalities | Manufacturing collapse

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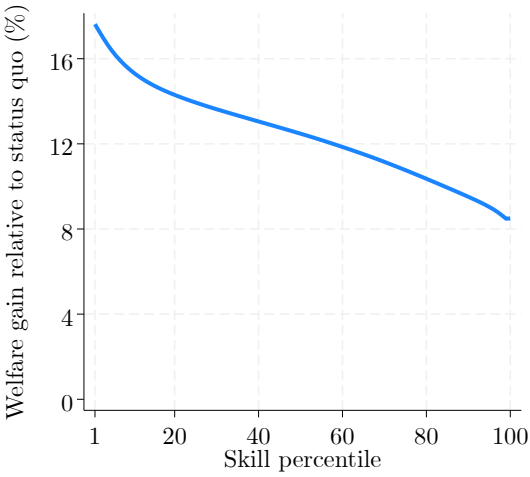
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- Gains **take time to materialize** and are **geographically spread out**.
  - → Local reforms might look disappointing despite **large aggregate/long-term gains**.

# Appendix

# Who benefits from upzoning?

# Removing zoning benefits low-income workers more

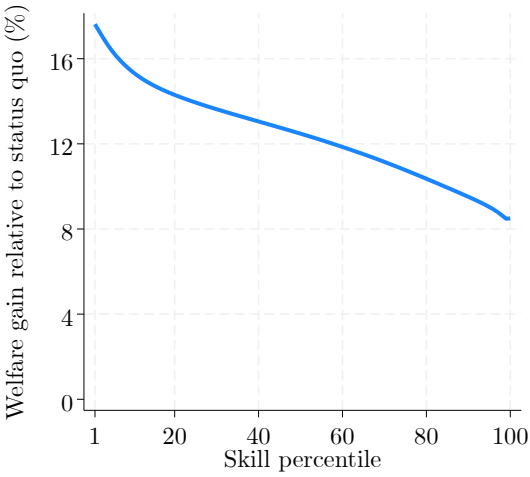
Back




- Lower-income workers spend a larger share of income on housing.

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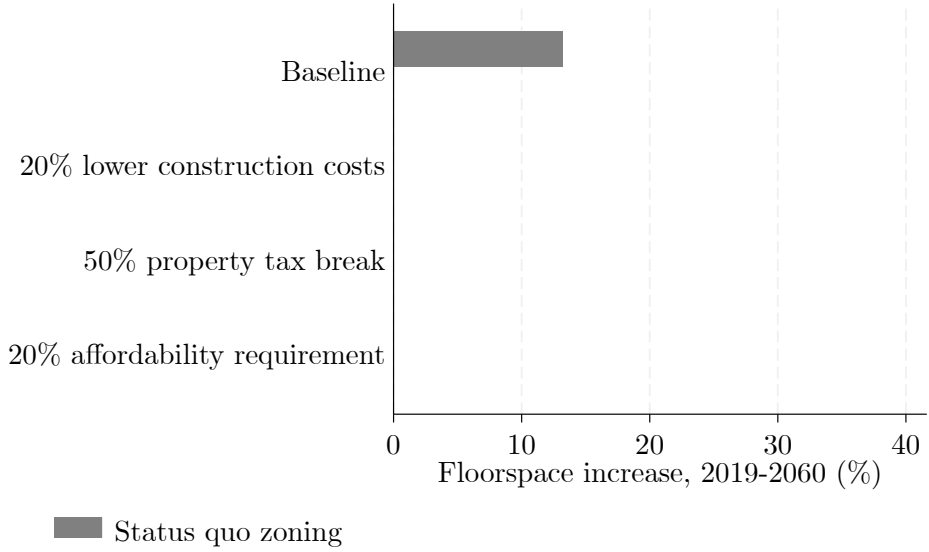
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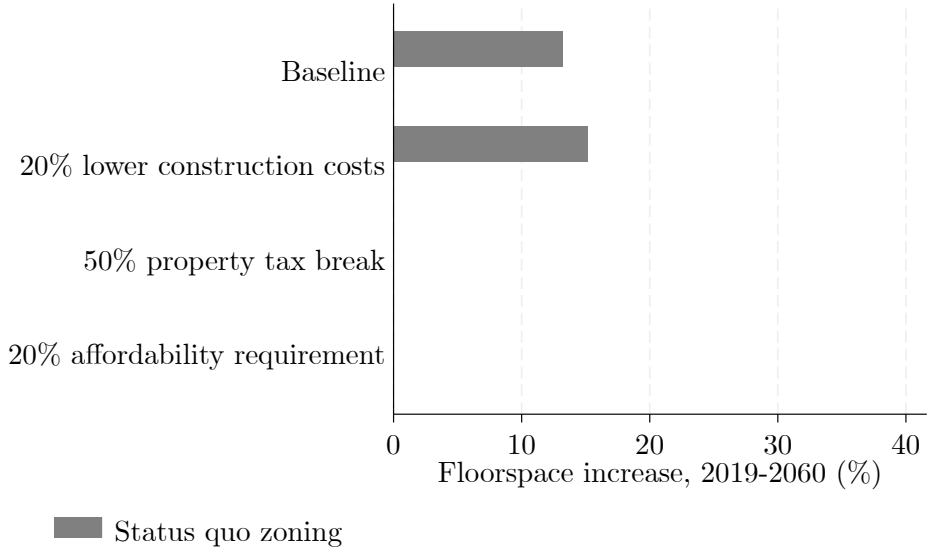

- Lower-income workers spend a larger share of income on housing.
- Increasing the housing stock lowers prices more in cheaper neighborhoods.

How effective are alternative policies to favor construction and affordability?

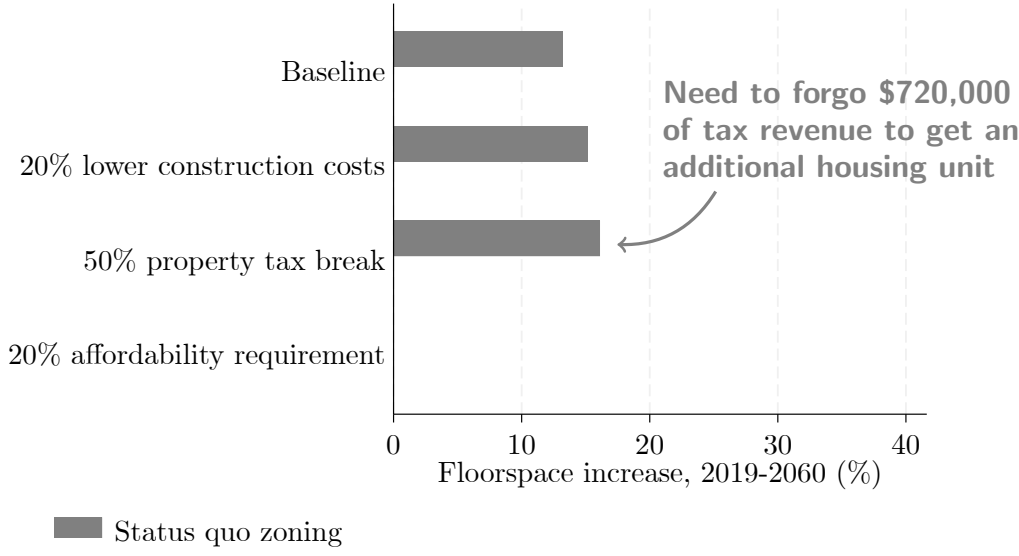
# Alternative policies to favor construction and affordability



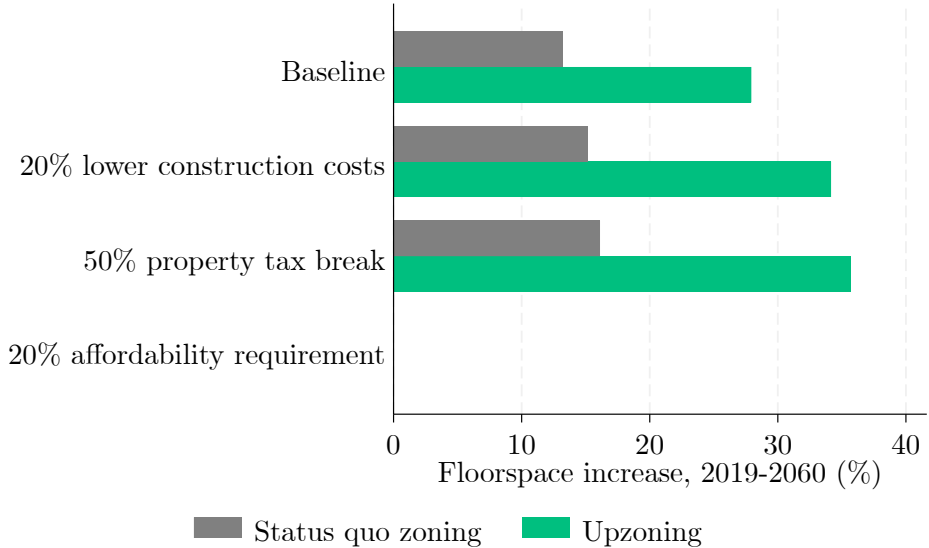
# Lowering construction costs marginally boosts construction



# Tax breaks on new buildings are relatively ineffective and costly

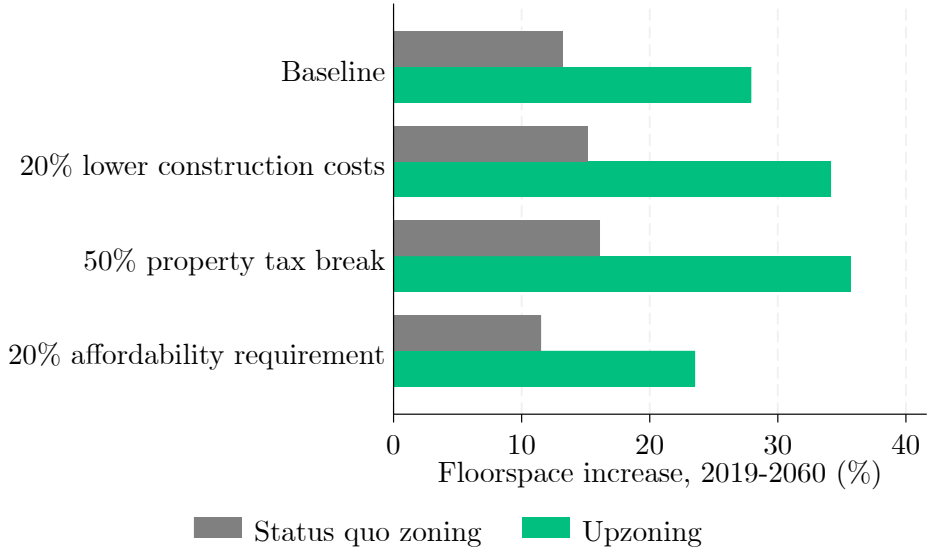


# Upzoning dominates alternative policies



# Inclusionary zoning: supply ↓ 0.6 sq.ft per affordable sq.ft

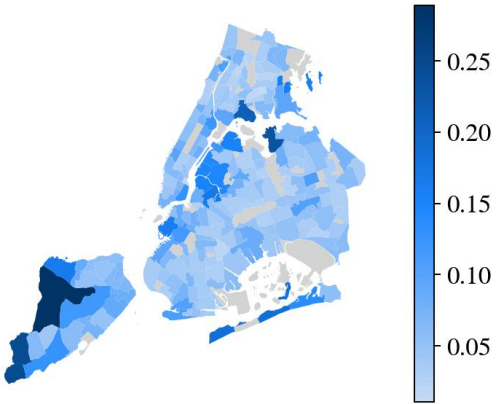
Back



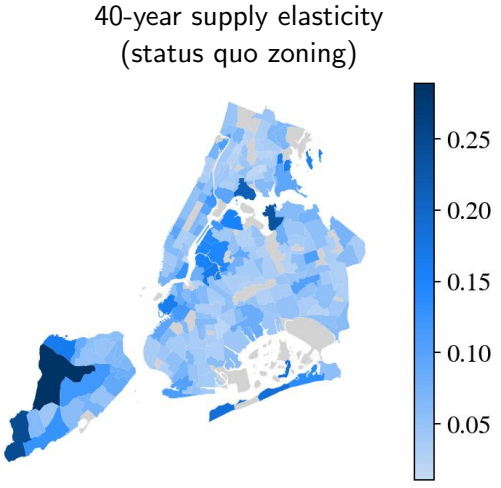
# How does zoning affect floorspace supply elasticities?

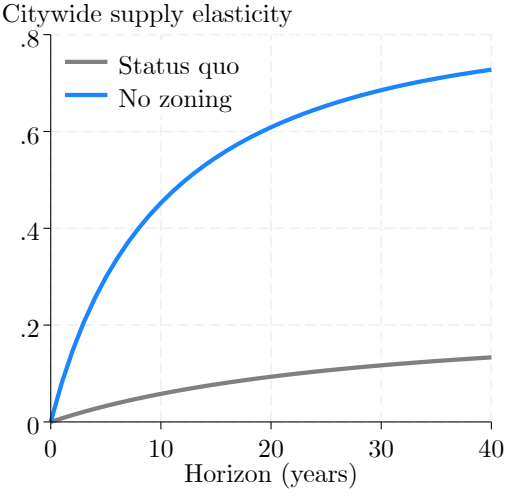
# Supply elasticities vary widely across neighborhoods

40-year supply elasticity
(status quo zoning)



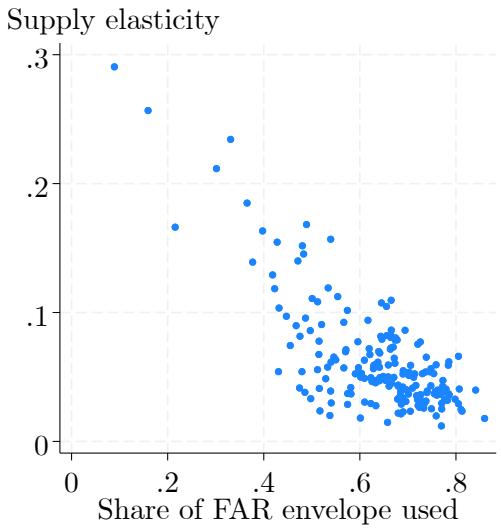
# Supply elasticities vary widely across neighborhoods



# Zoning is the strongest correlate of supply elasticities

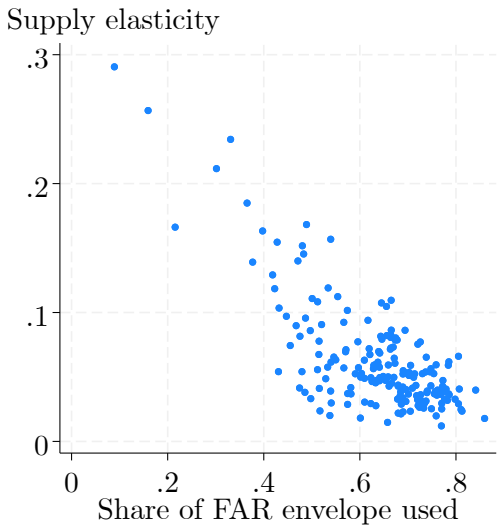
Back




- Supply elasticities are largely determined by the share of the zoning envelope that has already been built out.

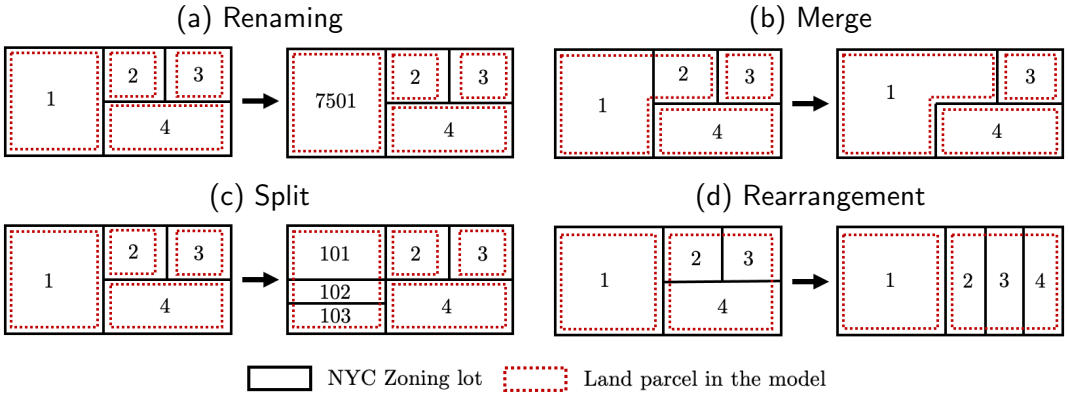
# Zoning is the strongest correlate of supply elasticities

Back



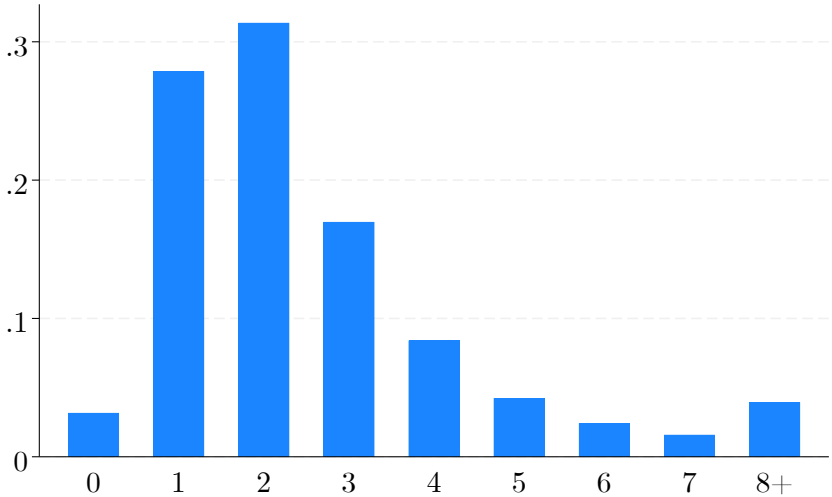

- Supply elasticities are largely determined by the share of the zoning envelope that has already been built out.
- Without zoning, citywide supply elasticity 5x higher.
  → Mostly determined by built density. More

# Boundary changes (21,700 detected over 2004-2022)



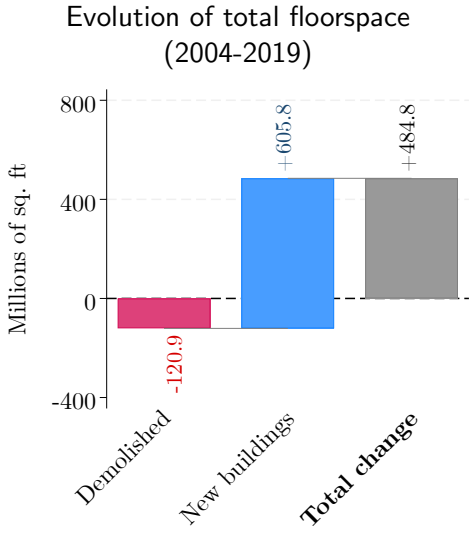

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# Redevelopment duration (in years)




Back

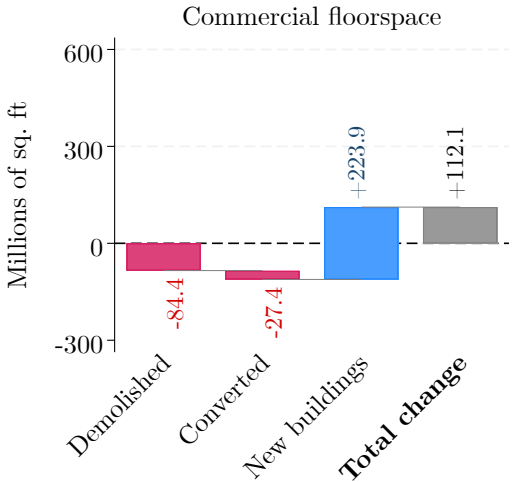
# Mature cities grow by redeveloping old structures

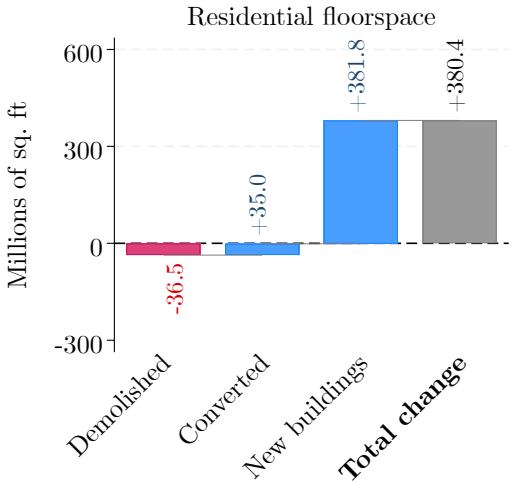



- Since 2004, floorspace in NYC has grown at a rate of $\sim$0.6% per year.

Back

# Changes in residential/commercial floorspace

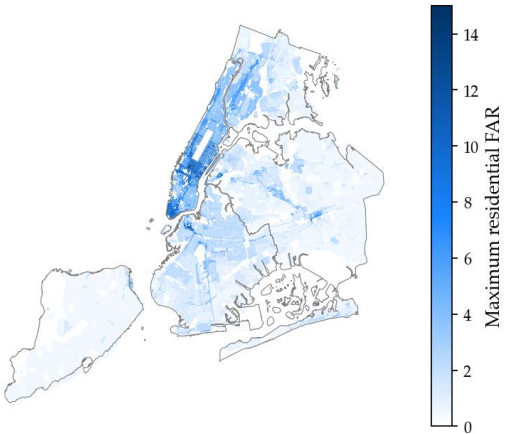


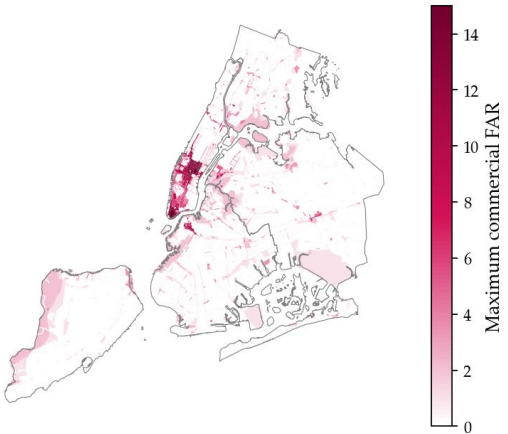

Back

# FAR allowances in NYC's zoning map

**Residential uses**

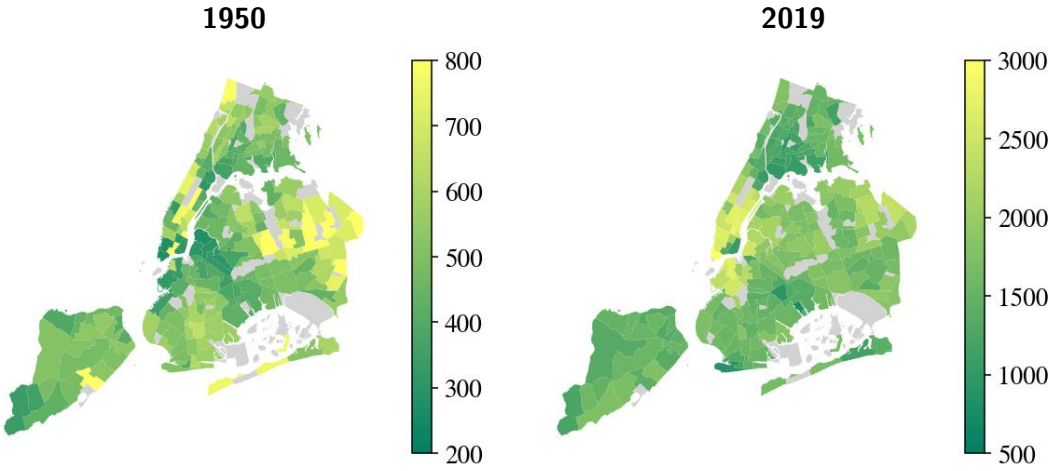



**Commercial uses**



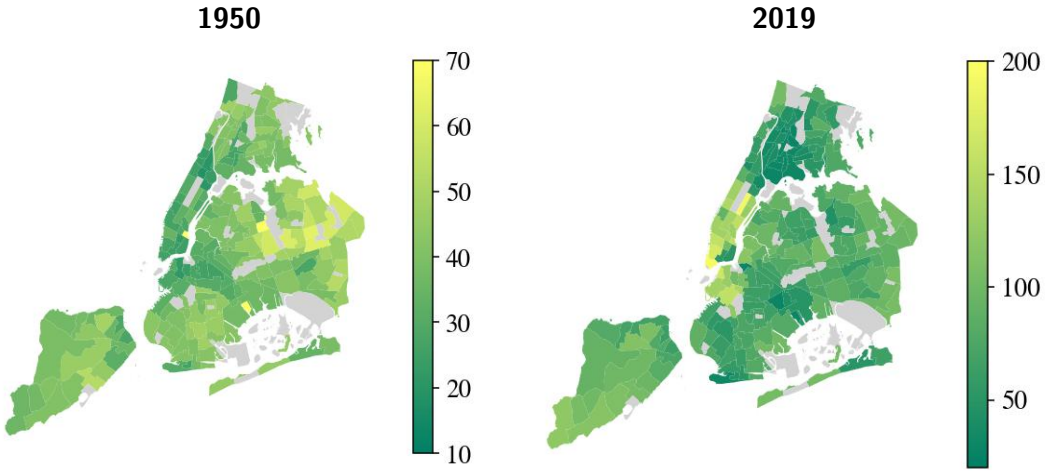

Back

# Evolution of median residential rents ($/month)



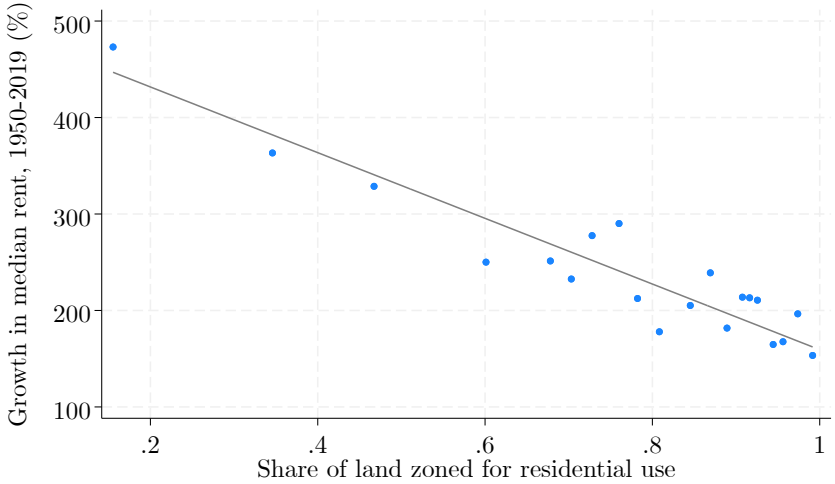

Back

# Evolution of median household income ($k/year)



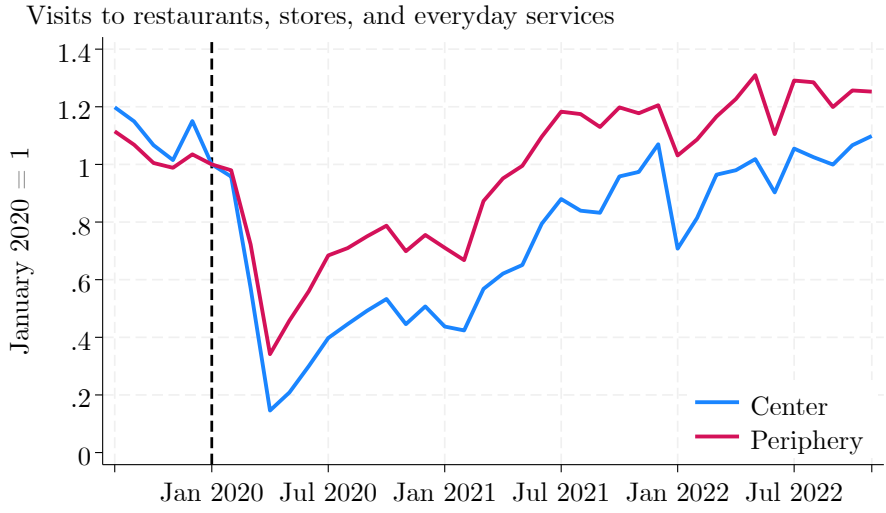

Back

# Commercial-oriented areas became more attractive over time



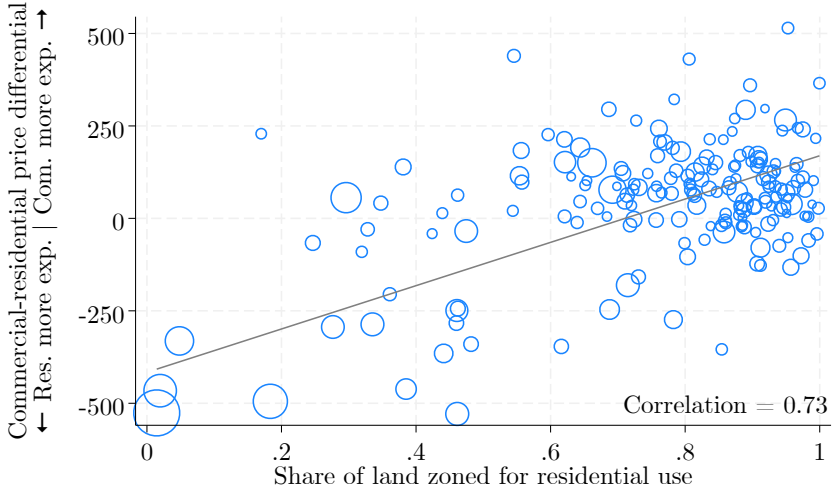

Back

# Work from home will likely accelerate existing trends




Back

# Use restrictions constrain the reallocation of land uses




Back

# Zoning's goals

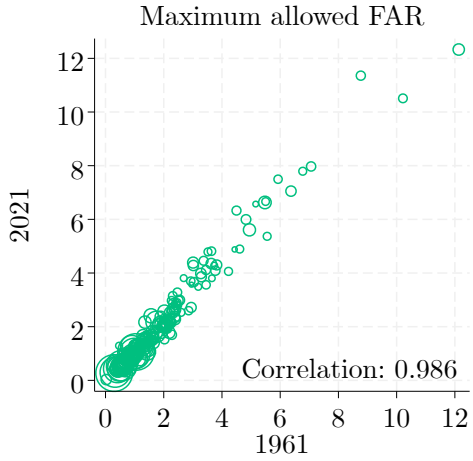
- Current zoning resolution adopted in 1961.
- Planners believed NYC has nearly reached its maximum size.
  → Didn't view the zoning code as restrictive.
- Aims of the 1961 zoning resolution:
  1. Promote tower-in-the-park development.
  2. Better separate commercial/residential uses.

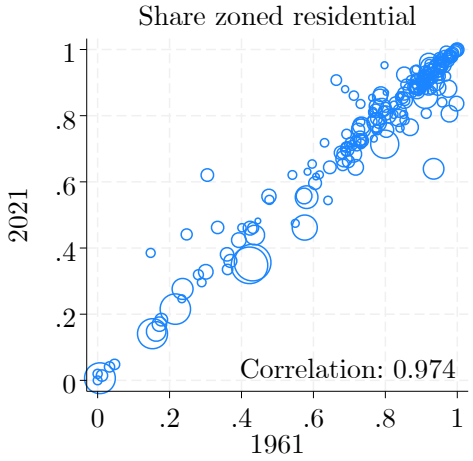
  → De facto: stabilization of existing land uses.
- Current zoning regulations closely align with the 1961 ordinance.



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# Since 1961: Persistence in zoning



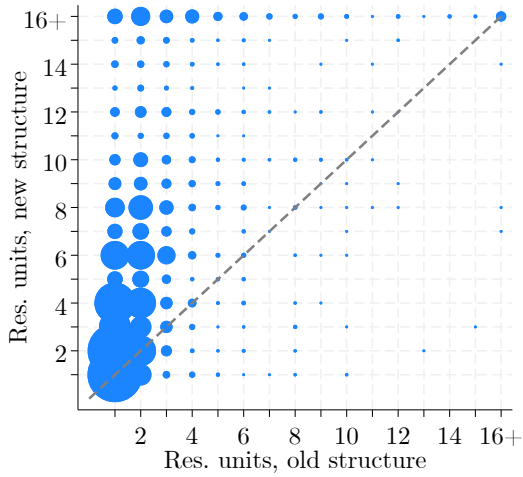

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# Change in number of units

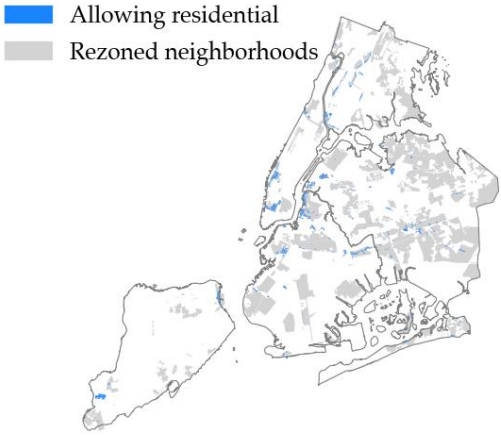
## Residential-to-residential redevelopment

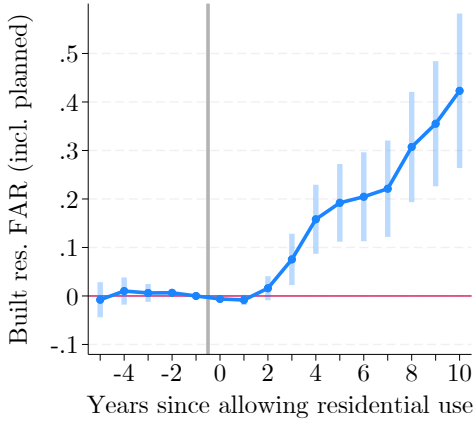
- New residential units are about 10% larger than the ones they replace.
- The number of units in new residential buildings is, on average, 3 times larger than in old structures.

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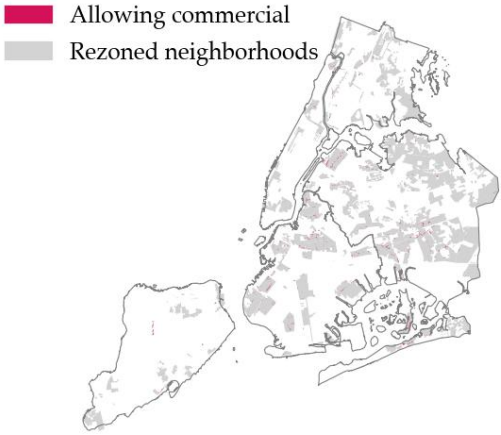
# Effects of allowing residential use

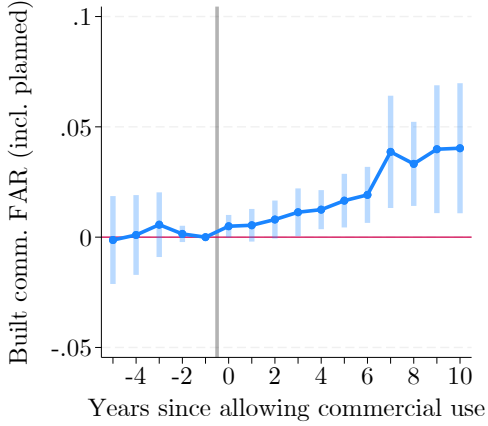



◂ Back (upzoning event study) ◂ Back (price differentials)

# Effects of allowing commercial use



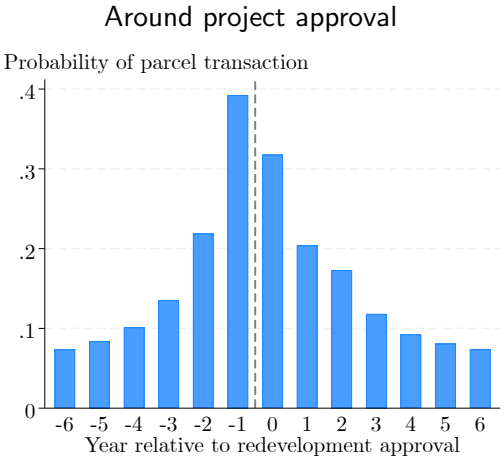

◂ Back (upzoning event study) ◂ Back (price differentials)

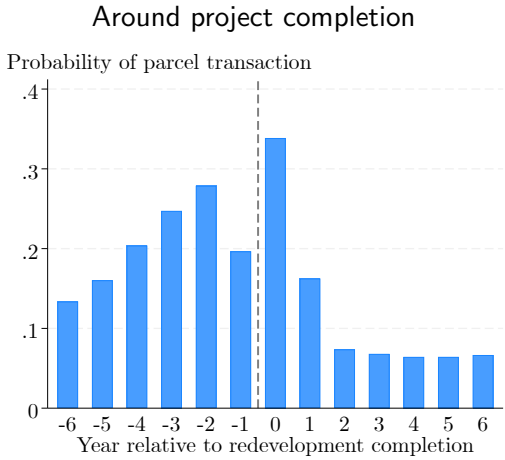
# Probability of a parcel sale (probit regression)

| | Probit coefficients | |
|---|---|---|
| Parcel sold | | |
| Office space (% of total floorspace) | -0.076 | (0.006) |
| Retail space (% of total floorspace) | -0.018 | (0.004) |
| Garage space (% of total floorspace) | 0.000 | (0.007) |
| Storage space (% of total floorspace) | 0.033 | (0.009) |
| Factory space (% of total floorspace) | 0.001 | (0.007) |
| Hotel space (% of total floorspace) | -0.015 | (0.020) |
| Other space (% of total floorspace) | -0.327 | (0.006) |
| Condo/Coop | -0.684 | (0.008) |
| Parcel in Bronx | 0.051 | (0.003) |
| Parcel in Brooklyn | 0.038 | (0.003) |
| Parcel in Queens | 0.036 | (0.003) |
| Parcel in Staten Island | 0.023 | (0.003) |
| Constant | -1.615 | (0.003) |
| Observations | 13,156,064 | |

Back

# Parcel sales around redevelopment events




Back

# Estimation details: Building values

- Using the sales data, I estimate separately for residential and commercial structures:

$$\log(\text{price}_s) = \underbrace{\rho^{\theta}_{1,n(s)}}_{\text{Neigh. FE}} + \underbrace{\rho^{\theta}_{2,bt(s)}}_{\text{Borough} \times \text{year FE}} + \boldsymbol{\beta} \underbrace{(\mathbf{x}_s - \bar{\mathbf{x}}_s)}_{\text{Structure characteristics}} + \nu_s$$

- Expected value of floorspace in a new building: $\bar{p}_{nt} = \exp(\hat{\rho}_{1,n} + \hat{\rho}_{2,b(n)t} + \hat{\sigma}_{\nu}/2)$
- Negative coefficient on (log) FAR, more negative for commercial.
  - → Less usable floorspace in tall buildings (e.g., because of mechanical space).
  - → Lower floor of commercial buildings is more valuable.

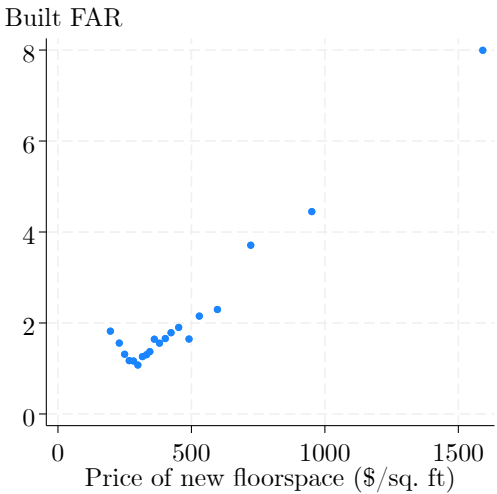
Back

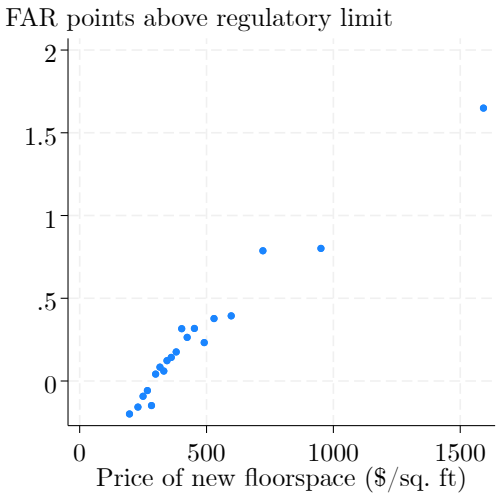
# Hedonic regression coefficients

Back

| | (1) Residential | | (2) Commercial | |
|---|---|---|---|---|
| (log) Built FAR | -0.027 | (0.002) | -0.091 | (0.011) |
| (log) Unit size | -0.065 | (0.002) | -0.108 | (0.005) |
| Age | -0.002 | (0.000) | 0.000 | (0.000) |
| Rent-stabilized | -0.336 | (0.002) | | |
| Landmark | -0.078 | (0.006) | 0.087 | (0.060) |
| Grade A | 0.162 | (0.003) | 0.222 | (0.032) |
| Grade B | 0.025 | (0.002) | 0.095 | (0.019) |
| Grade C | 0.007 | (0.002) | -0.024 | (0.019) |
| Brick | -0.155 | (0.003) | -0.001 | (0.084) |
| Frame | -0.068 | (0.003) | -0.224 | (0.118) |
| Masonry | -0.156 | (0.003) | -0.071 | (0.019) |
| Office building | | | 0.147 | (0.025) |
| Retail building | | | 0.222 | (0.021) |
| Garage building | | | 0.004 | (0.025) |
| Industrial building | | | 0.030 | (0.025) |
| Hotel | | | 0.352 | (0.048) |
| Neighborhood FE | Yes | | Yes | |
| Borough $\times$ Year FE | Yes | | Yes | |
| Observations | 428,338 | | 15,795 | |

# Developers build taller when facing high prices



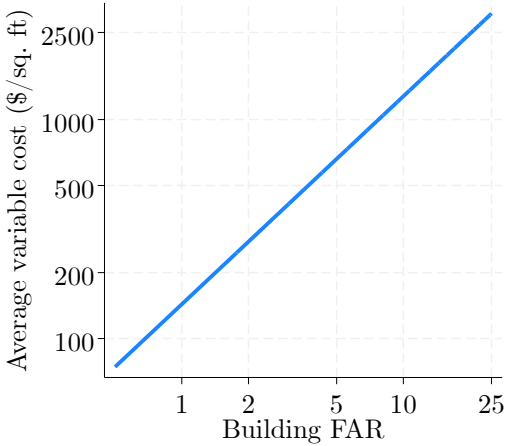

Back

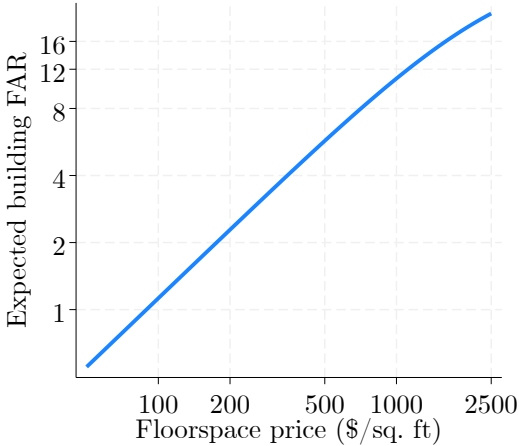
# Estimated cost parameters

| | | | |
|---|---|---|---|
| **(a) Variable cost parameters** | | | |
| $\alpha^0$ | Baseline cost of materials | 80.4 | (2.3) |
| $\zeta$ | Capital cost share | 0.51 | (0.005) |
| $\sigma_\eta$ | Cost shock standard deviation | 1.08 | (0.02) |
| **(b) Fixed cost parameters** | | | |
| $\delta^0$ | Base fixed cost | 175.2 | (7.7) |
| $\delta^{\text{demolition}}$ | Demolition multiplier | 1853.8 | (37.6) |
| $\delta^{\text{density}}$ | Neighborhood density multiplier | 318.5 | (11.3) |
| $\delta^{\text{protected}}$ | Protected parcels multiplier | 0.77 | (0.03) |
| **(c) Profit shock parameters** | | | |
| $\sigma_\epsilon^0$ | Base profit shock variance | 73.7 | (2.9) |
| $\sigma_\epsilon^{\text{demolition}}$ | Demolition multiplier | 138.8 | (5.1) |
| $\sigma_\epsilon^{\text{density}}$ | Neighborhood density multiplier | 248.2 | (4.9) |

◂ Back (variable costs) ◂ Back (fixed costs)

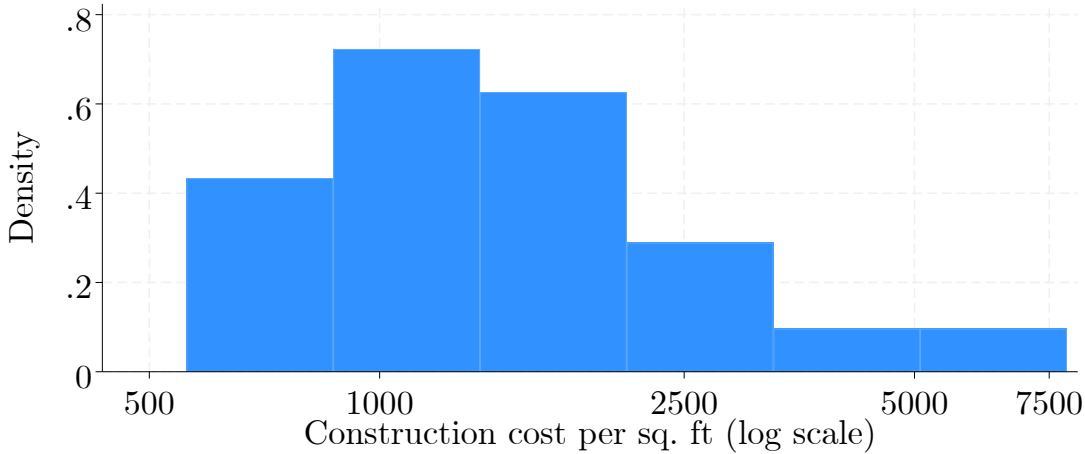
# Construction costs




Back

# Construction costs for skyscrapers




Back

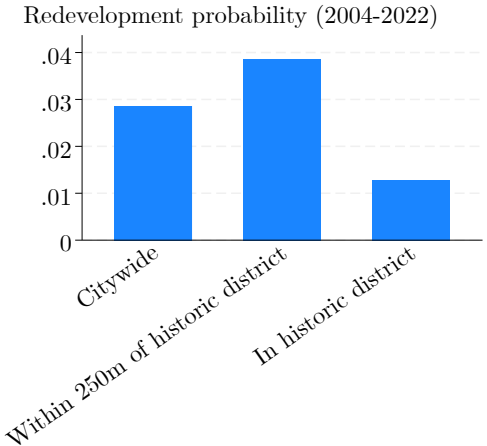
# Fixed costs parameterization

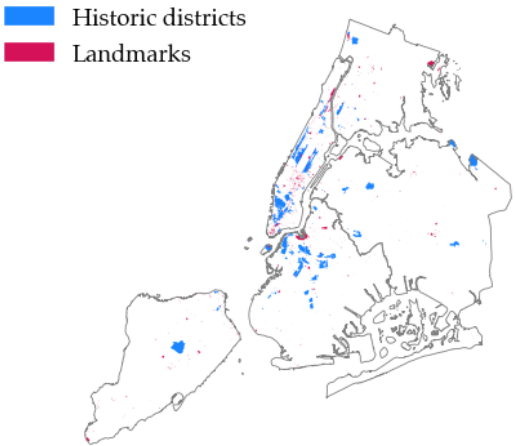
$$FC_{it} = (\delta^0 + \delta^{\text{demolition}} h_{it}^{\text{old}} + \delta^{\text{density}} \bar{h}_{n(i)t})(1 + \delta^{\text{protected}} \mathbf{1}_{it}^{\text{protected}}).$$

- $\delta^0$: Baseline fixed cost.
- $h_{it}^{\text{old}}$: FAR of building to demolish.
- $\bar{h}_{n(i)t}$: Average neighborhood density.
- $\mathbf{1}_{it}^{\text{protected}}$: Landmarked parcel or in historic district.

Back

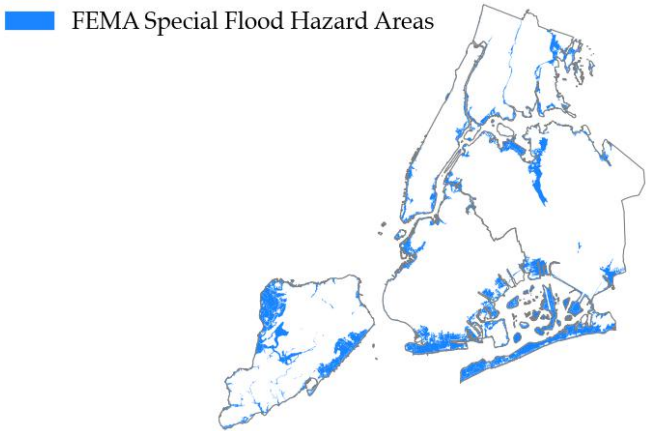
# Protected areas




◂ Back (fixed costs) ◂ Back (counterfactuals)

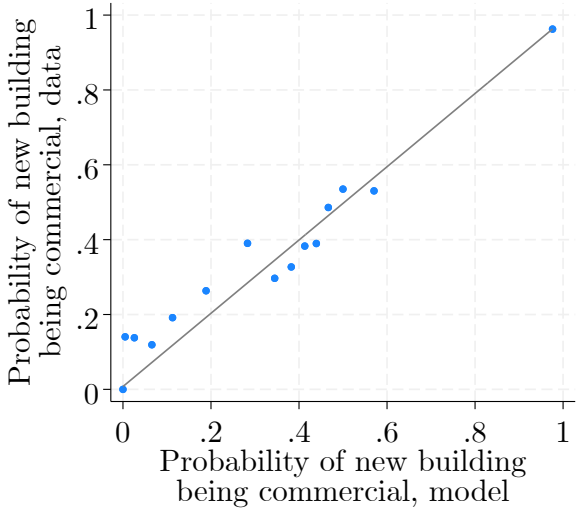
# Flood zones



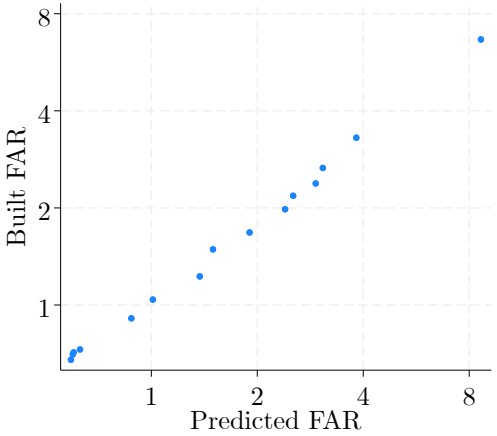

Back

# Model fit: Residential vs. commercial construction

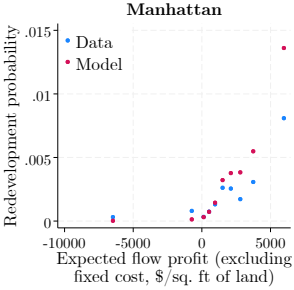
Back

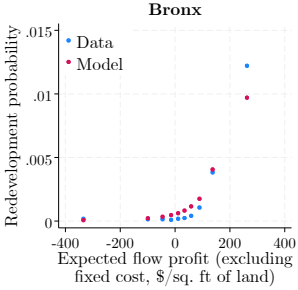


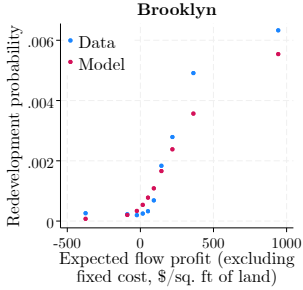
# Fit on extensive and intensive margins

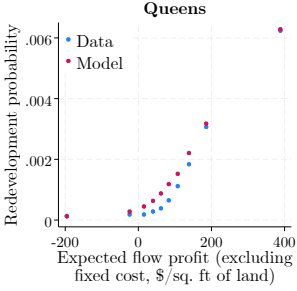


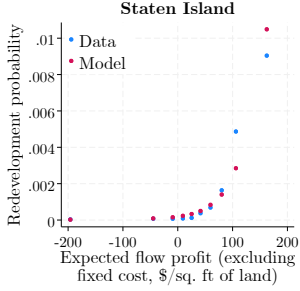

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Manhattan
Data
Model
Redevelopment probability
.015
.01
.005
0
-10000
-5000
0
5000
Expected flow profit (excluding fixed cost, $/sq. ft of land)


Bronx
Data
Model
Redevelopment probability
.015
.01
.005
0
-400
-200
0
200
400
Expected flow profit (excluding fixed cost, $/sq. ft of land)


Brooklyn
Data
Model
Redevelopment probability
.006
.004
.002
0
-500
0
500
1000
Expected flow profit (excluding fixed cost, $/sq. ft of land)


Queens
Data
Model
Redevelopment probability
.006
.004
.002
0
-200
0
200
400
Expected flow profit (excluding fixed cost, $/sq. ft of land)


Staten Island
Data
Model
Redevelopment probability
.01
.008
.006
.004
.002
0
-200
-100
0
100
200
Expected flow profit (excluding fixed cost, $/sq. ft of land)

Back

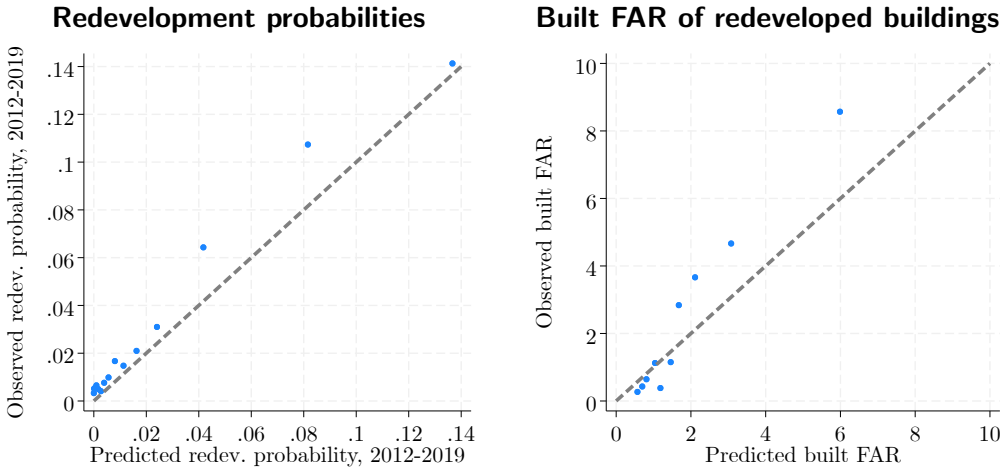
# Out-of-sample model fit

Back

- Re-estimate the model using first half of the data (2004–2011):
  - → Price levels;
  - → Preferences and location fundamentals;
  - → Sale probabilities;
  - → Variable costs;
  - → Fixed costs.
- Predict the evolution of the city over 2012–2019.

# Out-of-sample model fit: Parcel level

Back

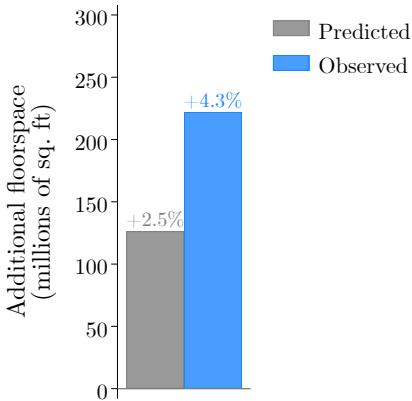



→ Differences explained by high price growth post-2012.

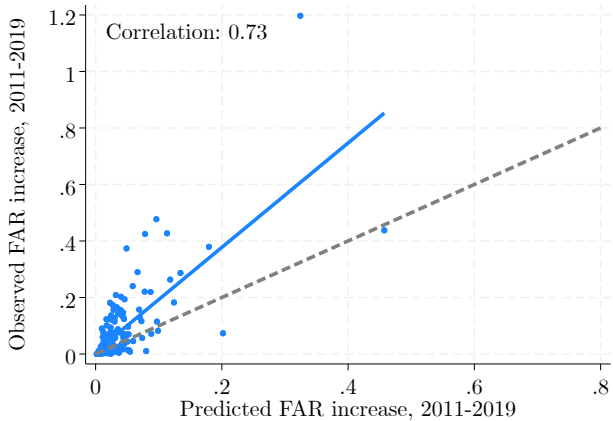
# Out-of-sample model fit: Aggregate growth

Back

**Citywide floorspace growth**




**Neighborhood-level growth**




→ Differences explained by high price growth post-2012.

# Demand model estimation (1)

- Calibrate $\beta$ to 0.1 using the ACS.
- Estimate the shape of $z^W$ at 4.4 using the commuting data.
- Calibrate wages from the number of people working in each location.
  - → High-wage locations attract more workers, and from further away.
- Calibrate amenities $\boldsymbol{B}$ with residential prices and the number of residents in each location.
  - → Locations attracting many residents despite high prices must have high amenities.
- Calibrate subsistence levels $\bar{\boldsymbol{h}}$ with the total housing consumption in each location.
  - → Higher levels of $\bar{h}$ lead to more housing consumption.
  - → Average $\bar{h}$ of 224 sq. ft (IQR = [170 sq. ft, 265 sq. ft])

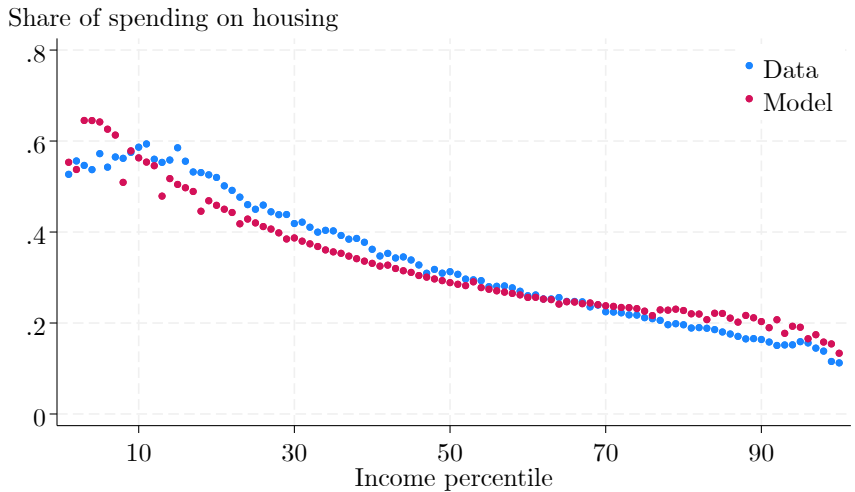
# Demand model estimation (2)

- Calibrate the shape of $z^H$ to 2.9 match the variance of average neighborhood incomes.
  - $\rightarrow$ Higher variance of $z^H \implies$ less sorting across neighborhood by income.

- Calibrate productivities $\boldsymbol{A}$ and floorspace shares in production $\boldsymbol{\alpha}$ using data on commercial floorspace quantities and prices, the number of jobs in each location, and the calibrated wages.
  - $\rightarrow$ Productivity is estimated to be higher near the center of the city.
  - $\rightarrow$ $\alpha$ averages 0.18 across neighborhoods (IQR $= [0.14, 0.21]$). Close to benchmarks in the literature (0.2 in Ahlfeldt et al., 2015; 0.16 in Greenwood, Hercowitz, and Krusell, 1997).

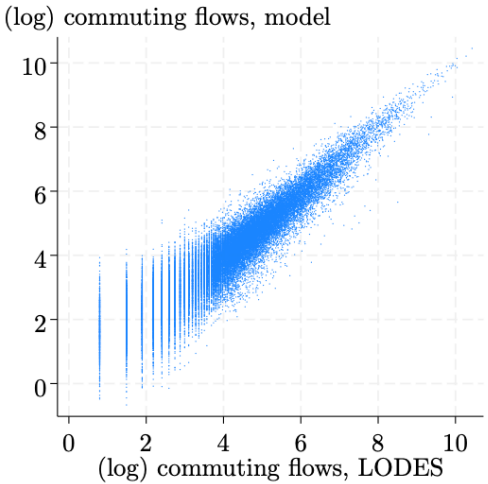
Back

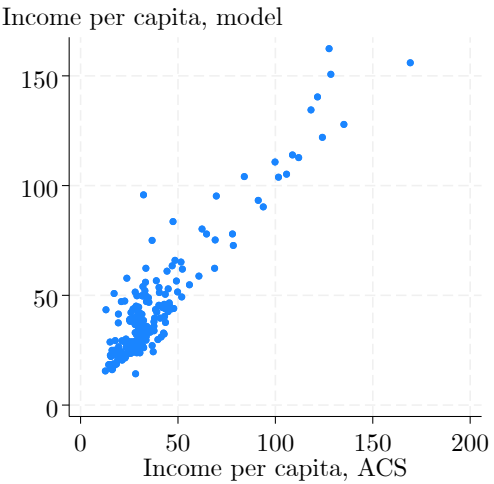
# Engel curve for housing




Back

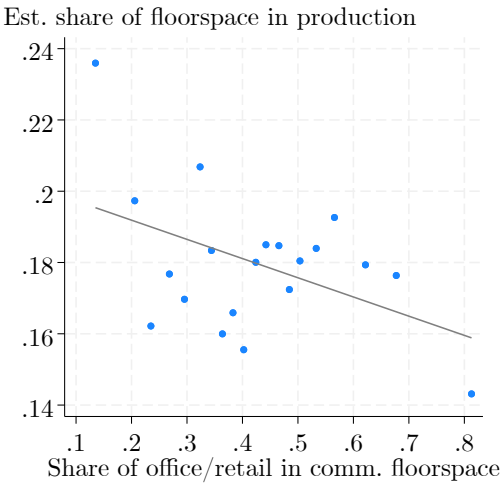
# Demand model fit: Commuting flows and sorting

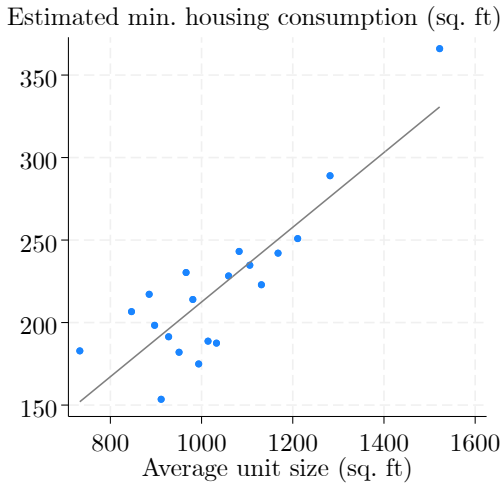



Back

# Demand model fit: $\alpha$ and $\bar{h}$




Back

| Amenity | Correlation between share of satisfied residents and $B$ |
|---|---|
| Neighborhood cleanliness | 0.47 |
| Control of street noise | 0.07 |
| Household garbage pick-up | 0.33 |
| Recycling services | 0.34 |
| Snow removal | 0.77 |
| Rat control | 0.36 |
| Bike safety | 0.25 |
| Pedestrian safety | 0.53 |
| Street maintenance | 0.53 |
| Parking enforcement | 0.74 |
| Storm water drainage and sewer maintenance | 0.57 |
| Availability of healthcare services | 0.52 |
| Availability of cultural activities | 0.60 |
| Neighborhood parks | 0.86 |
| Fire protection services | 0.75 |
| Emergency medical services | 0.73 |
| Neighborhood public safety | 0.66 |
| Bus services | 0.66 |
| Subway services | 0.71 |
| Public services | 0.63 |

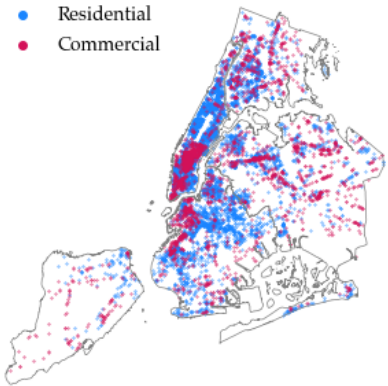
Back

# Large construction events

Construction events

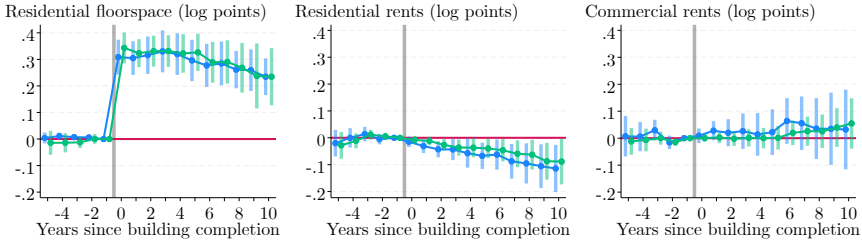



Buildings with rent data

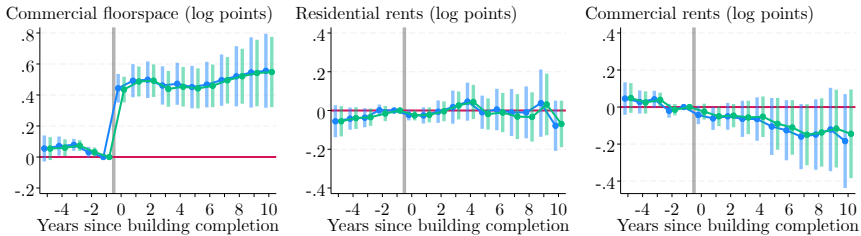



Back

## Effects of residential construction



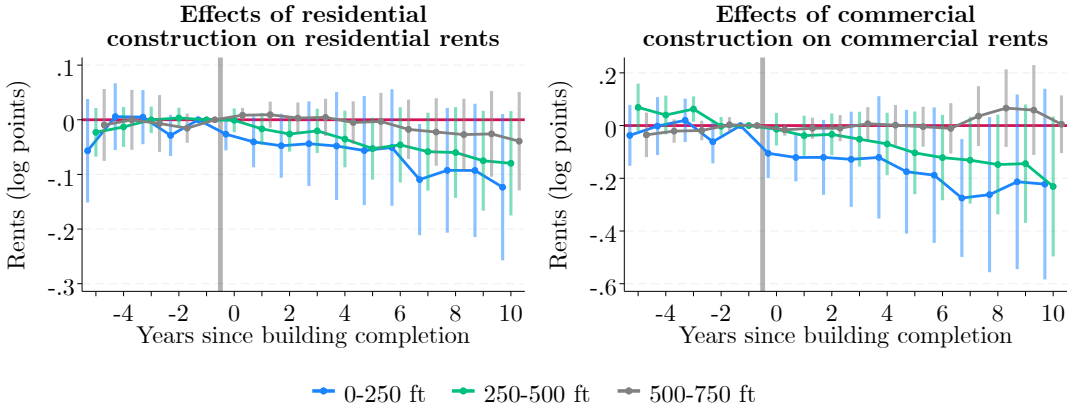

## Effects of commercial construction




Baseline With overlapping events

Back

# Spatial decay of price effects

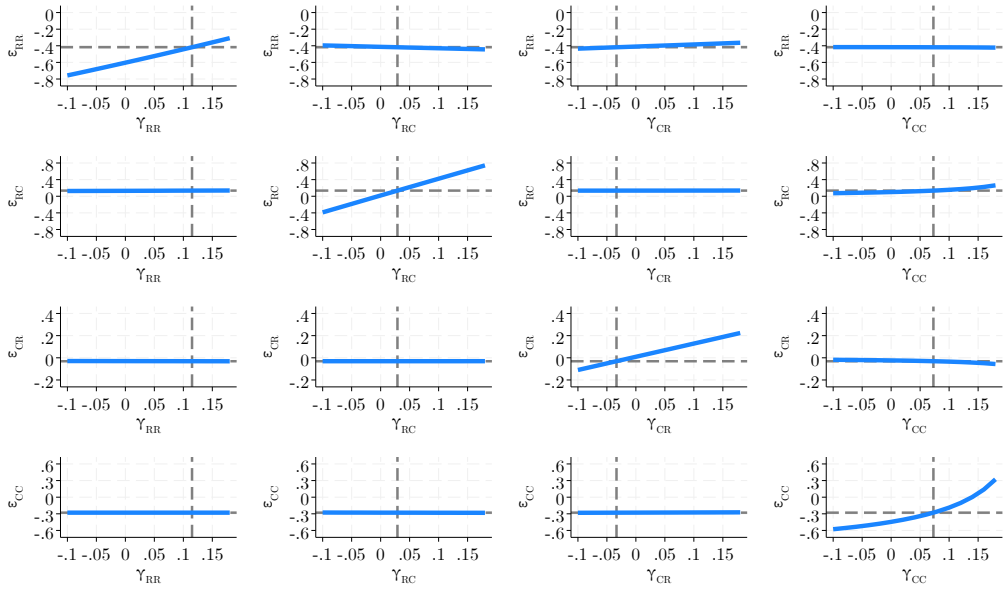



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# Estimated spillover parameters

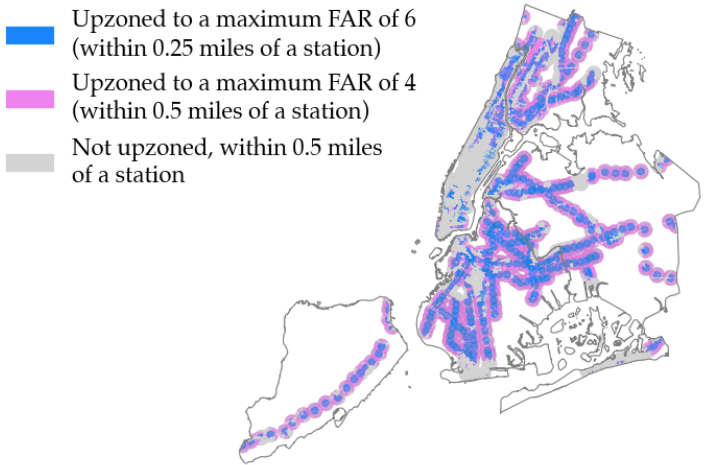
| Parameter | Interpretation | Calibrated value | Targeted elasticity |
|---|---|---|---|
| $\gamma^{RR}$ | Effect of residents on amenities | 0.11 | $\varepsilon^{RR} =$ -0.42 |
| $\gamma^{RC}$ | Effect of residents on productivity | 0.03 | $\varepsilon^{RC} = 0.14$ |
| $\gamma^{CR}$ | Effect of jobs on amenities | -0.03 | $\varepsilon^{CR} =$ -0.03 |
| $\gamma^{CC}$ | Effect of jobs on productivity | 0.07 | $\varepsilon^{CC} =$ -0.28 |

Back


$\varepsilon_{RR}$
$\varepsilon_{RC}$
$\varepsilon_{CR}$
$\varepsilon_{CC}$
$\gamma_{RR}$
$\gamma_{RC}$
$\gamma_{CR}$
$\gamma_{CC}$

Back

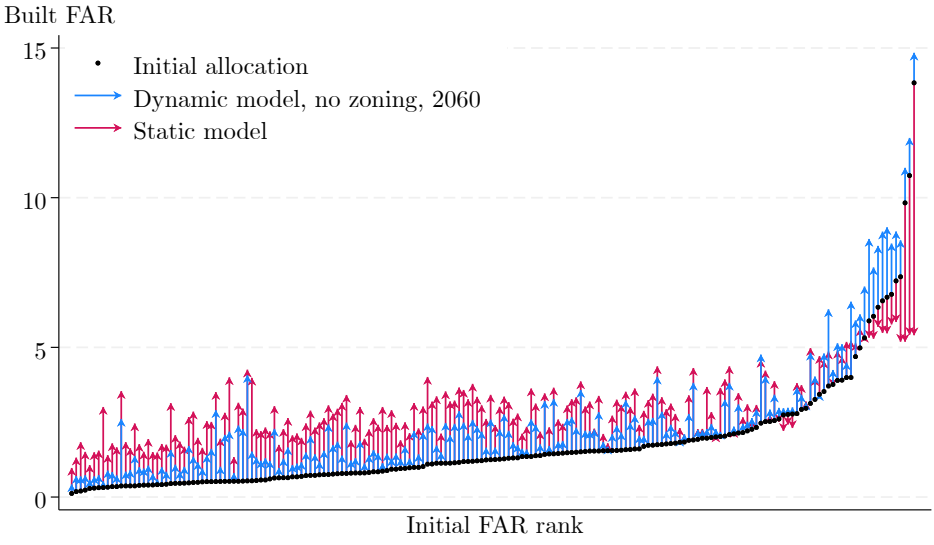
# Transit-Oriented Development




→ Upzones 37% of the city, mostly in outer boroughs.

Back

# Dynamic vs. static model

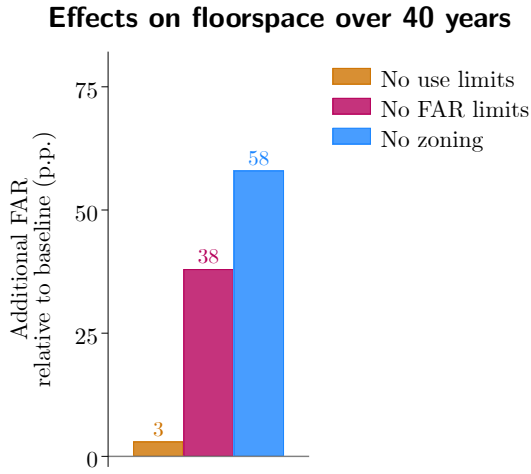



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# FAR vs. use limits

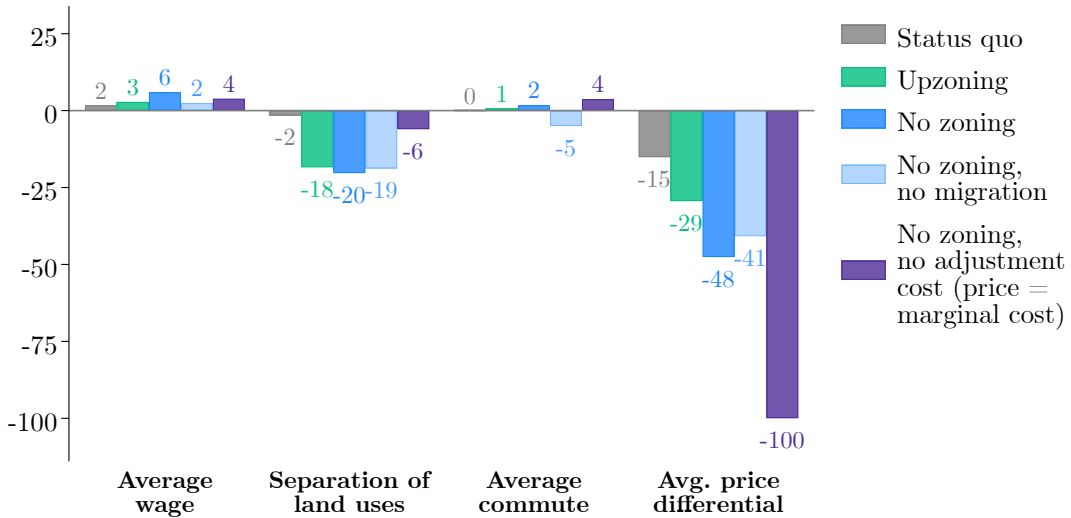
Back

- Relaxing use limits does not spur much redevelopment.
  - → Need to relax FAR limits to make redevelopment profitable.
- FAR and use limits interact.



# Additional outcomes

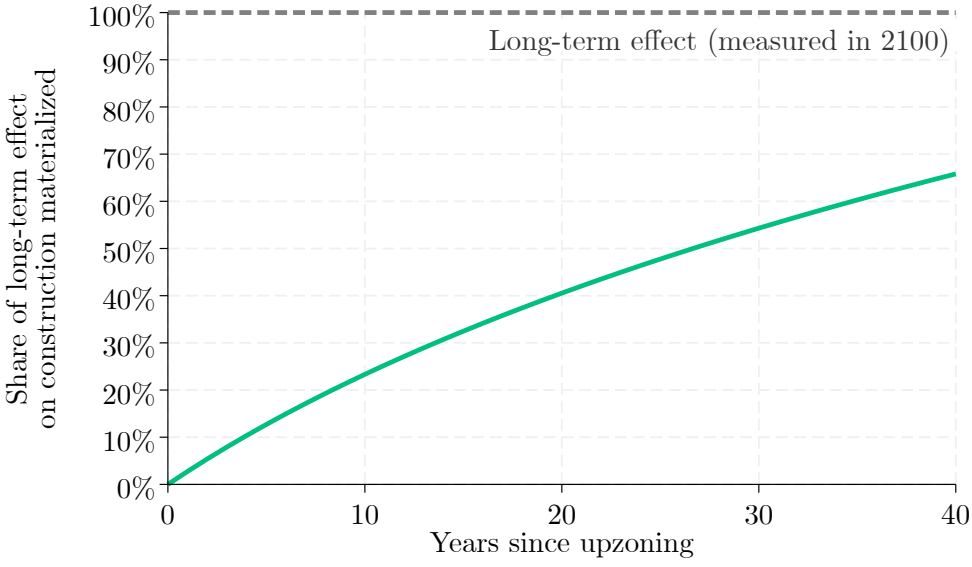
Change by 2060, relative to 2019 (%)




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# Effects of upzoning materialize slowly over time

Back



# Use vs. FAR limits

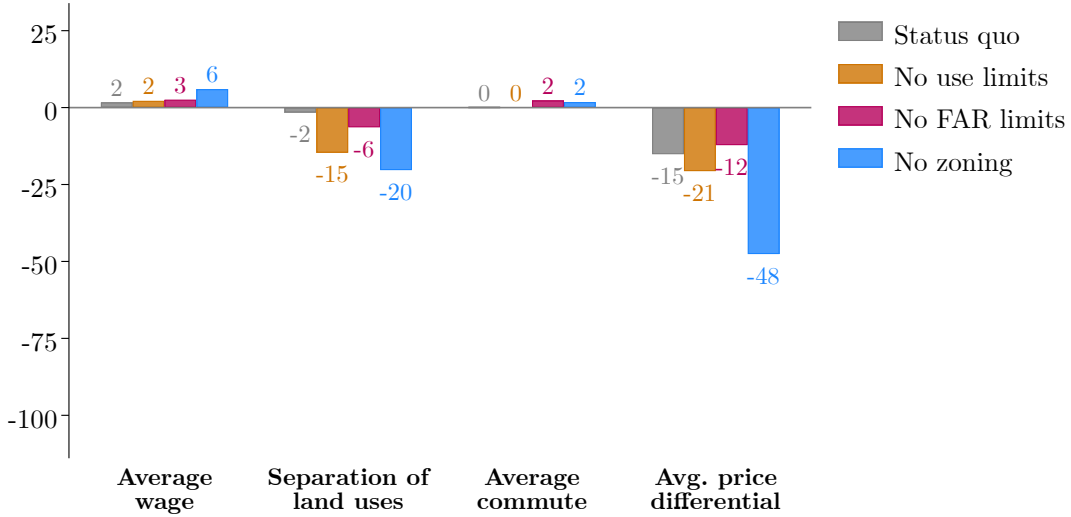
Change by 2060, relative to 2019 (%)




Back

# Use vs. FAR limits

Change by 2060, relative to 2019 (%)




Back

# No spillovers

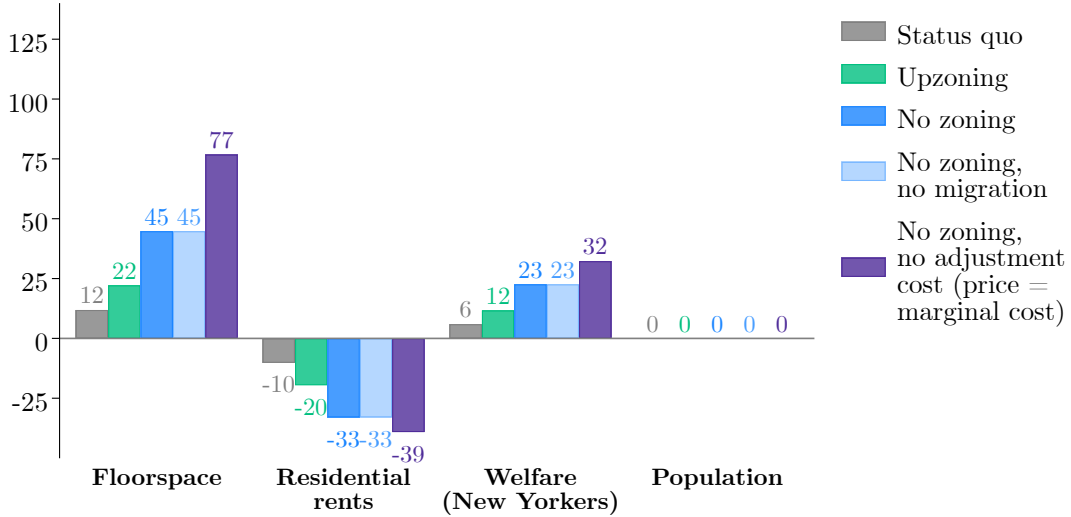
Change by 2060, relative to 2019 (%)




Back

# No migration

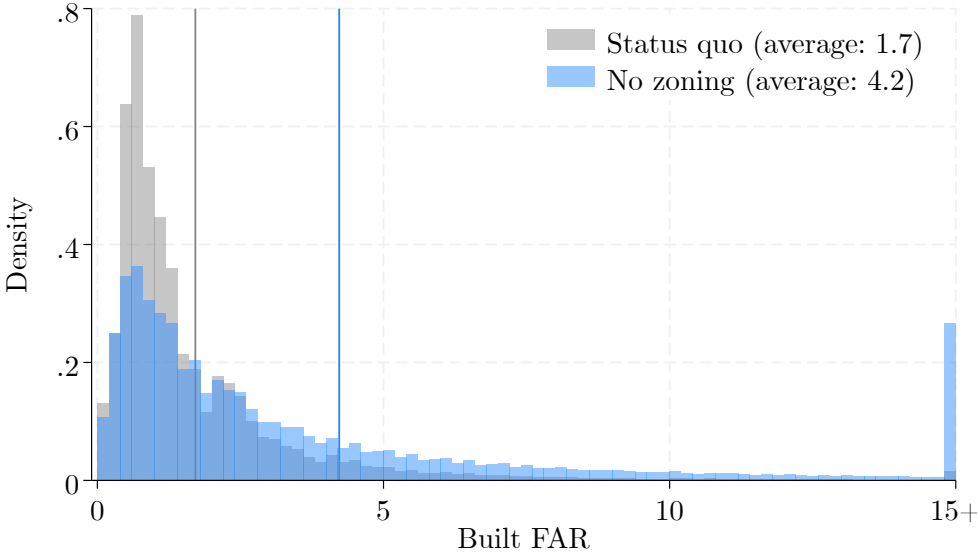
Change by 2060, relative to 2019 (%)




Back

# FAR distribution of new buildings

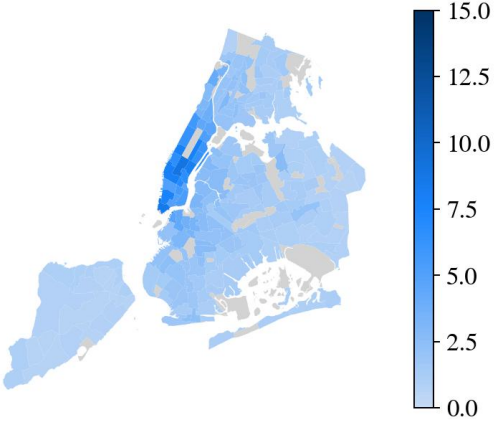
Back



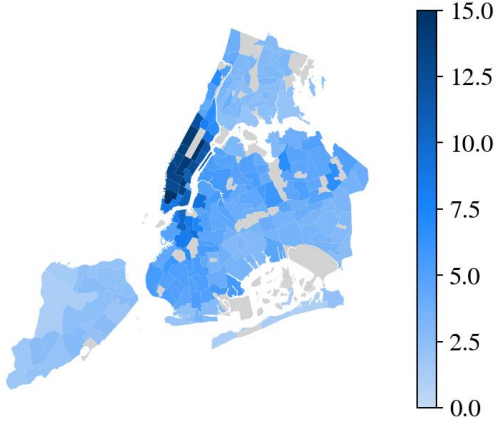
# Average FAR of new buildings

Back

**(a)** Status quo

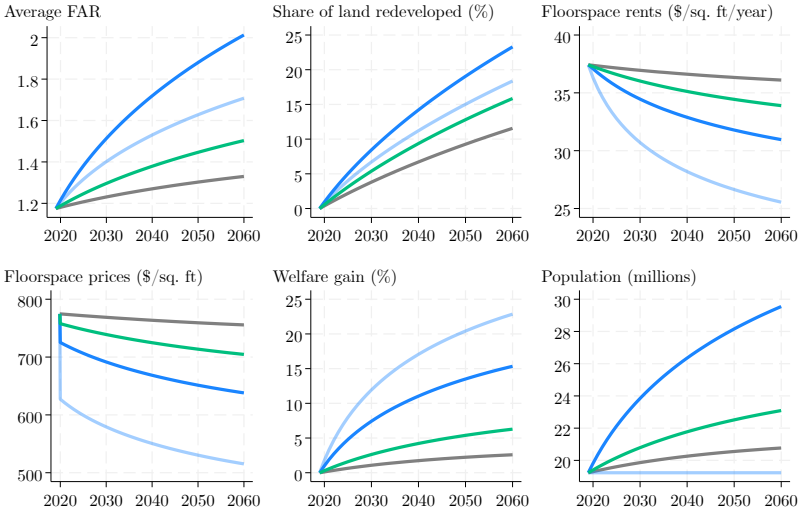



**(b)** No zoning



# Transition paths

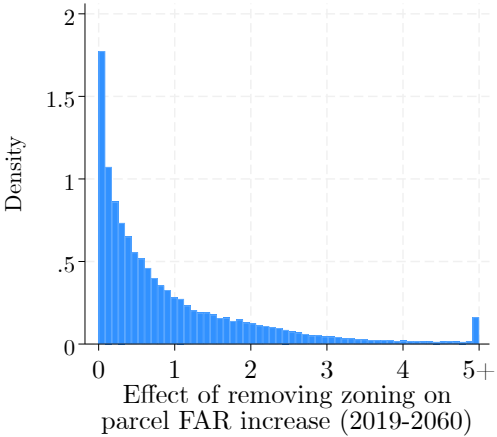
Back



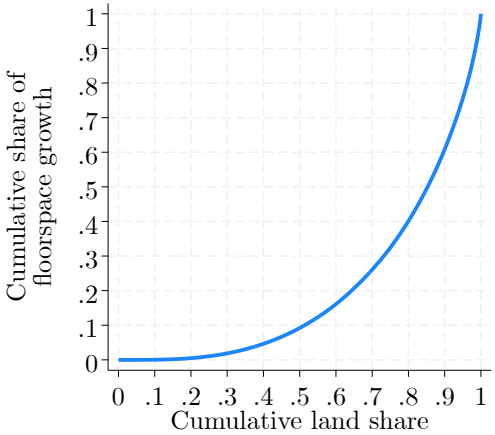
# Upzonings' effects are concentrated

Back

**Parcel-level distribution of FAR increases**

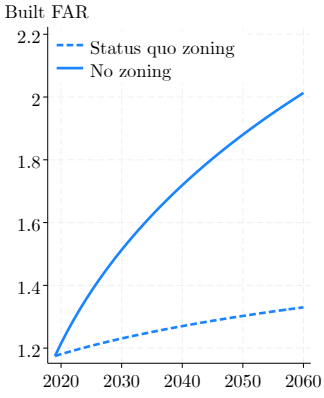



**Spatial concentration of floorspace growth**

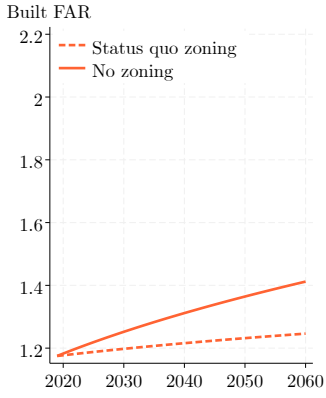


# Effects of the aggregate price level

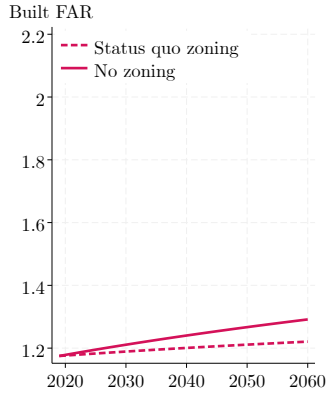
**Current price level**




**50% lower price level (Miami)**



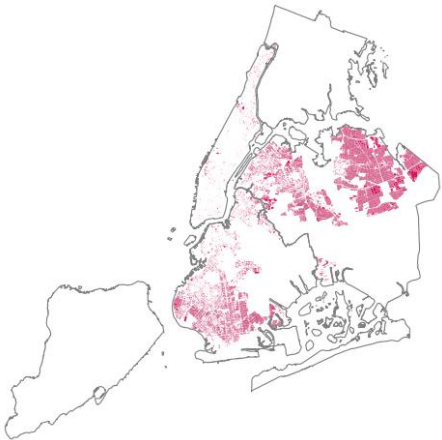

**67% lower price level (Chicago)**



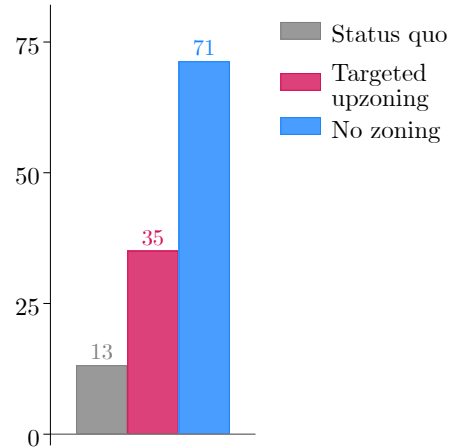

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# Targeted upzoning

**Upzoned parcels**



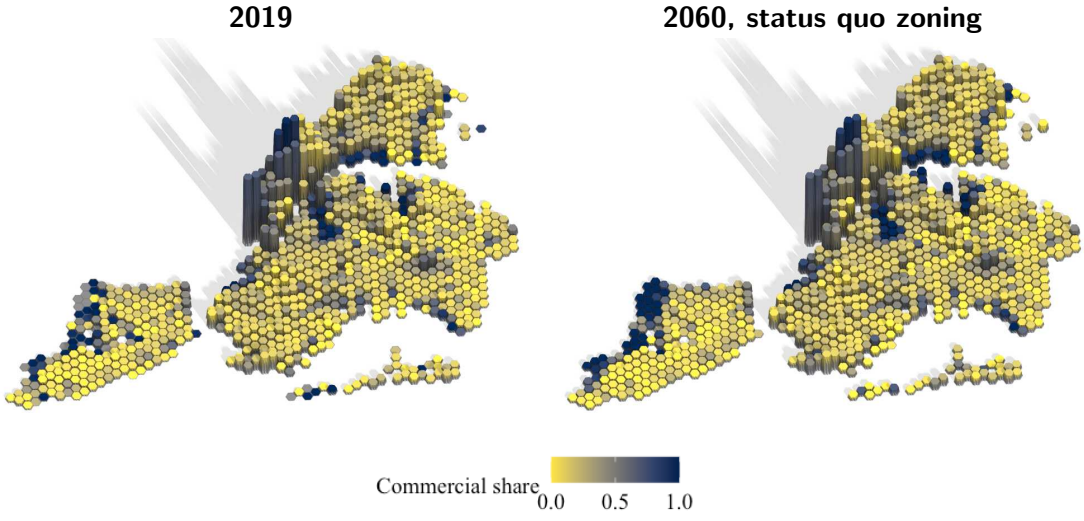
**Floorspace growth, 2019–2060**




Back

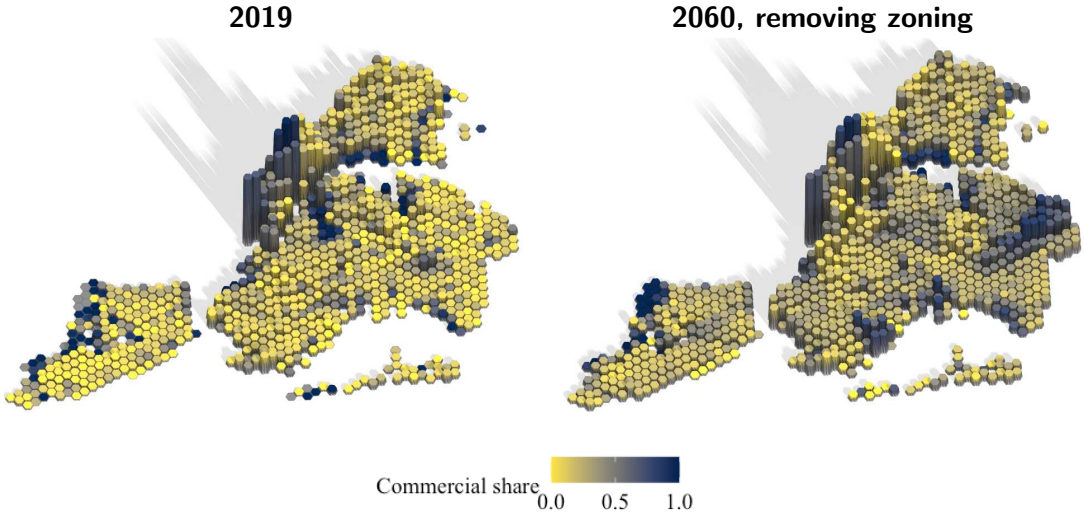
# Effects on city structure

Back



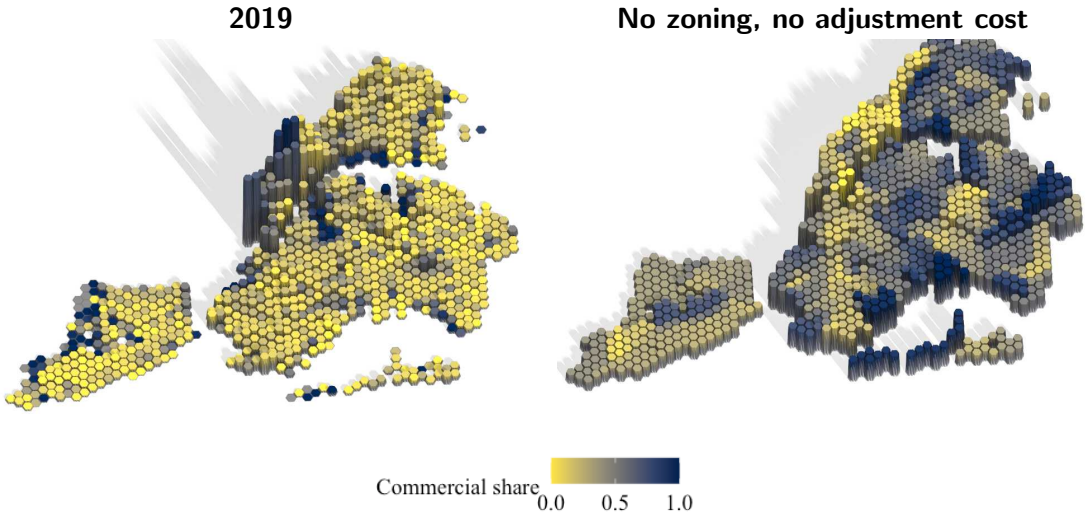
# Effects on city structure

Back

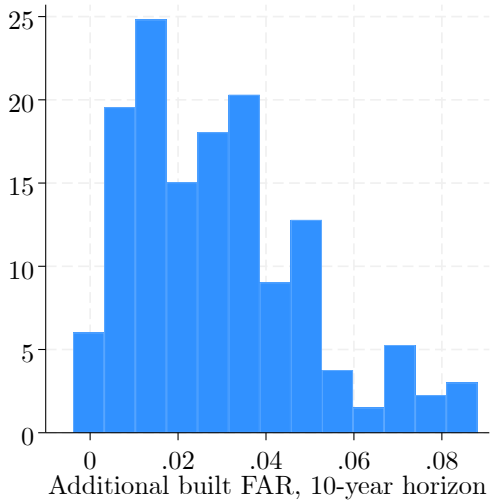


# Effects on city structure

Back



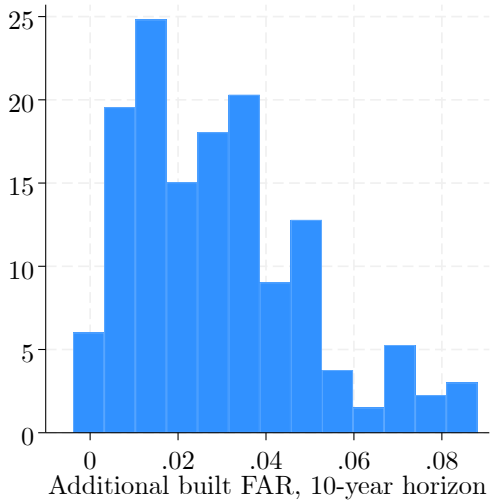
# Effects of upzoning vary widely across neighborhoods




- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
  → For comparison with event study.

Back

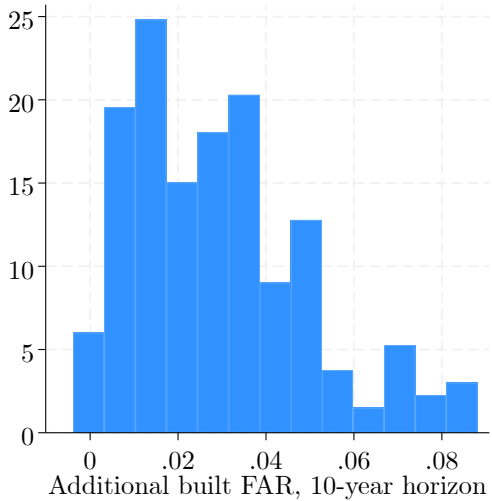
# Effects of upzoning vary widely across neighborhoods




- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
  - → For comparison with event study.
- Wide heterogeneity in effects.
  - → Important consequences if upzoning is politically costly.

Back

# Effects of upzoning vary widely across neighborhoods

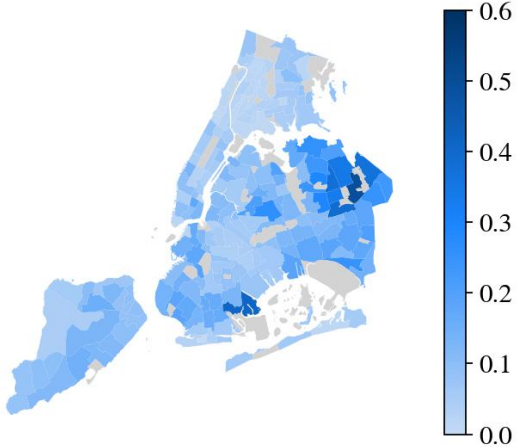


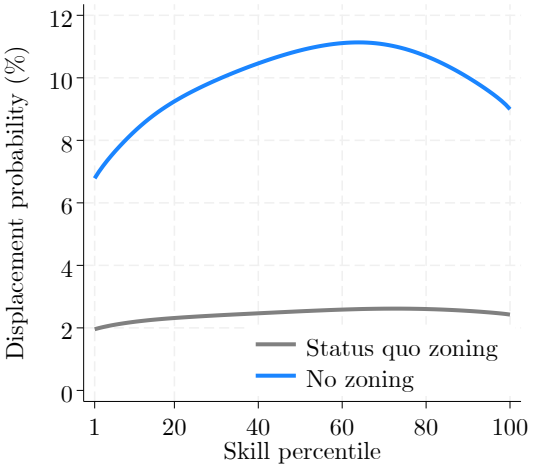

- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
  - → For comparison with event study.
- Wide heterogeneity in effects.
  - → Important consequences if upzoning is politically costly.
- Realized upzonings were in areas ripe for redevelopment.
  - → They were "selected on gains."

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# Richer households are more exposed to redevelopment

Share of land redeveloped by 2060
(no zoning regulations)

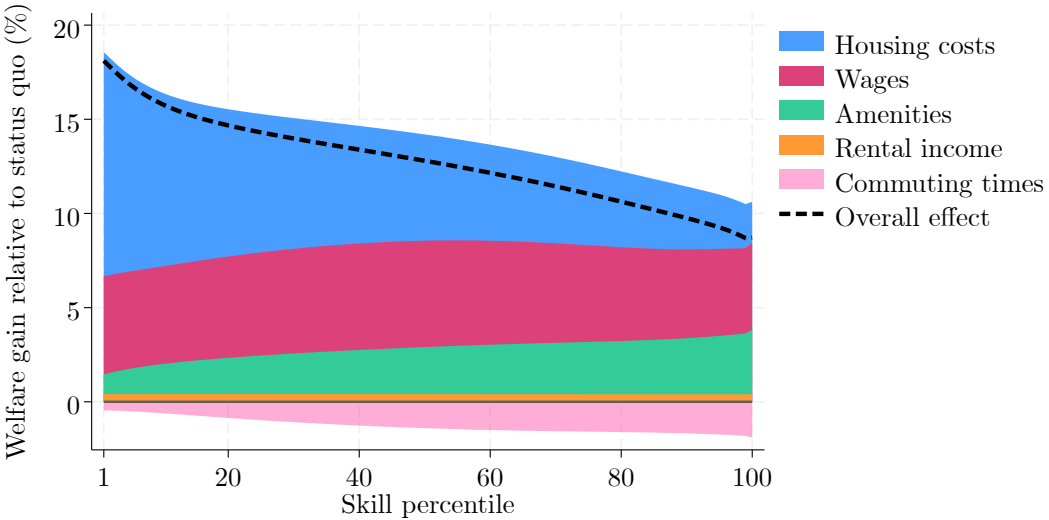


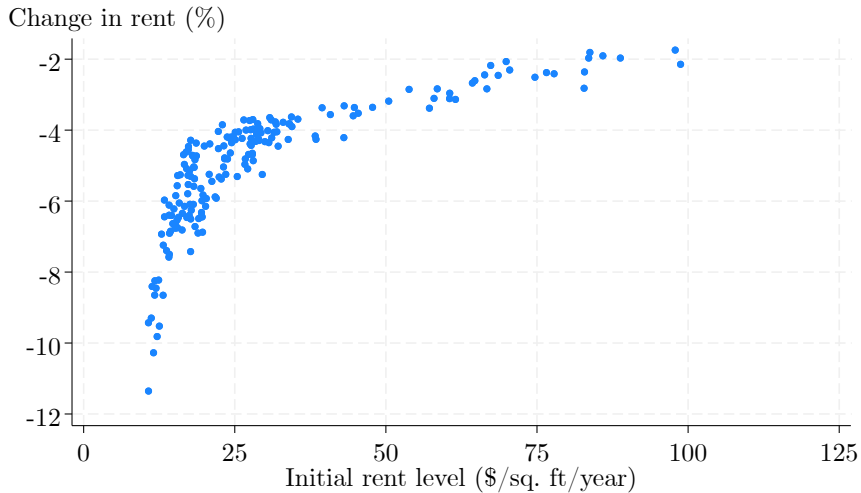

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# Decomposition of the welfare gains from removing zoning

Back

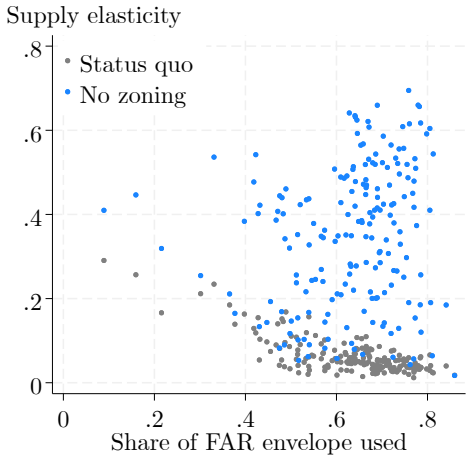


# Effects of a 10% increase in the city's floorspace on rents

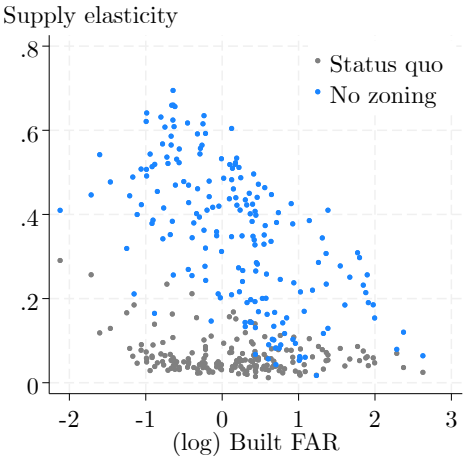



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# Determinants of supply elasticities




Correlation (status quo): -0.75
Correlation (no zoning): 0.16

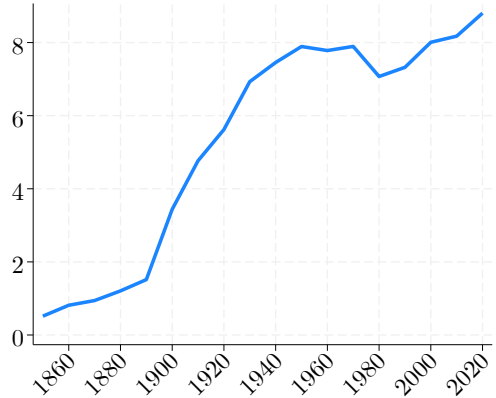



Correlation (status quo): -0.30
Correlation (no zoning): -0.61

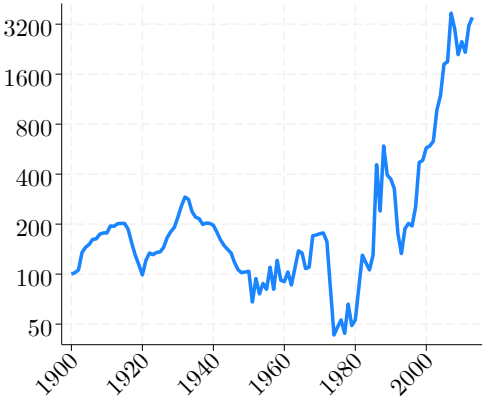
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# Historical population and land values

**(a)** NYC population (millions)

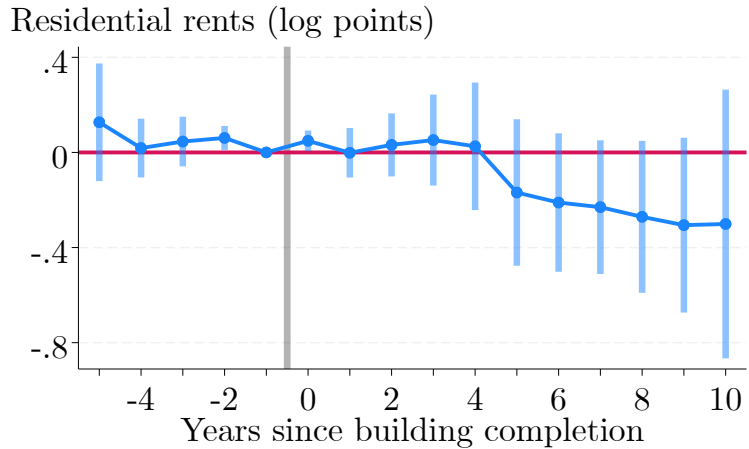



**(b)** Manhattan land values (1900 = 100)



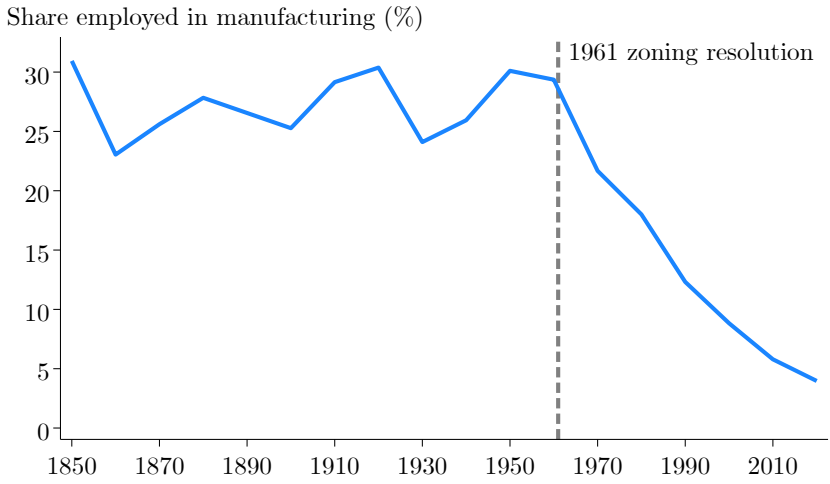

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# Effects of industrial construction on residential rents




◂ Back (Spillovers) ◂ Back (Why zoning)

# Manufacturing in NYC has collapsed since 1961




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